

Navigating with Stability: Local Minima, Patterns, and Evolution in a Gradient Damage Fracture Model

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Abstract

In phase-field theories of brittle fracture, crack initiation, growth and path selection are investigated using non-convex energy functionals and a stability criterion. The lack of convexity with respect to the state poses difficulties to monolithic solvers that aim to solve for kinematic and internal variables, simultaneously. In this paper, we inquire into the effectiveness of quasi-Newton algorithms as an alternative to conventional Newton-Raphson solvers. These algorithms improve convergence by constructing a positive definite approximation of the Hessian, bargaining improved convergence with the risk of missing bifurcation points and stability thresholds. Our study focuses on one-dimensional phase-field fracture models of brittle thin films on elastic foundations. Within this framework, in the absence of irreversibility constraint, we construct an equilibrium map that represents all stable and unstable equilibrium states as a function of the external load, using well-known branch-following bifurcation techniques. Our main finding is that quasi-Newton algorithms fail to select stable evolution paths without exact second variation information. To solve this issue, we perform a spectral analysis of the full Hessian, providing optimal perturbations that enable quasi-Newton methods to follow a stable and potentially unique path for crack evolution. Finally, we discuss the stability issues and optimal perturbations in the case when the damage irreversibility is present, changing the topological structure of the set of admissible perturbations from a linear vector space to a convex cone.

1 Introduction

Numerous physical phenomena in materials science such as crystal plasticity, phase transitions, twinning [19], and fracture [50, 6, 7], can be described by non-linear energy functionals at the mesoscale. The configurational variables within these energy functionals evolve under external loading, navigating equilibrium states. These states corresponds to critical points of the energy functional, satisfying both boundary conditions and an optimality criterion. This optimization process is achieved through incremental minimization along a loading program. Outcomes of such optimization are fields (e.g., displacement, strain, stress), energy components, and order parameters depending on the model considered. The corresponding microstructures are crucial to understand and improve the mechanical behavior of materials.

Functionals of the type $\Psi(u)$, non-convex in their argument u (a displacement field) are frequently employed in theories such as quasi-continuum methods, the multi-well Landau-type theory of weak or reconstructive phase transformations, twinning, and crystal plasticity [77, 49, 21, 27, 19, 69, 3, 5]. On the other hand, a second type of functionals, denoted as $\Psi(u, \alpha)$, find application in phase-field theories. Here, the scalar phase-field variable α is an internal variable that elucidates the substance's state, encompassing aspects like crystal structure, symmetry, lattice orientation, [27, 65, 39] or serving as a damage parameter in the variational phase-field theory of fracture [50, 67].

In both cases, one deals with the problem of finding configurations that satisfy $\min_u \Psi(u)$, $\min_{u, \alpha} \Psi(u, \alpha)$ or at least *some* necessary conditions for energy optimality and, potentially, accounting for constraints. In the nonconvex setting, multiple minima of the functionals and the multitude of equilibrium states

Description	Symbol	Remarks
Energy stiff	Ψ	
Energy compliant	$\tilde{\Psi}$	
Load	$\bar{\epsilon}_t$	
Load parameter	$t > 0$	
Homogeneous energy, stiff	Ψ^{hom}	
Homogeneous energy, compliant	$\tilde{\Psi}^{\text{hom}}$	
Material functions	$E(\alpha), w(\alpha)$	
Internal damage length	$\bar{\ell}$	dimensional, [m]
Film thickness	h	dimensional, [m]
Characteristic size	L	dimensional, [m]
Young modulus, substrate	E'_{2d}	
Young modulus, film	E_{2d}	
State, stiff	(u, α)	
State, compliant	(u, α, v)	
Equilibrium state, stiff	$y_t = (u_t, \alpha_t)$	
Equilibrium state, compliant	$y_t = (u_t, \alpha_t, v_t)$	
Homogeneous equilibrium state	y^{hom}	
State, perturbations (stiff)	$(u, \alpha), (w, \beta)$	
State, perturbations (compliant)	$(u, \alpha, v), (w, \beta, \tilde{v})$	
Admissible perturbations, stiff	V	$H^1(0, 1) \times H^1(0, 1)$
Admissible perturbations, compliant	$\tilde{V}$	$H^1(0, 1) \times H^1(0, 1) \times H^1(0, 1)$
State space, stiff	X_t	$H_t^1(0, 1) \times H^1(0, 1)$
State space, compliant	$\tilde{X}_t$	$H_t^1(0, 1) \times H^1(0, 1)$
Homogeneous space	V_0	$H_0^1(0, 1) \times H^1(0, 1)$
Cone of admissible perturbations	K_0^+	$H_0^1(0, 1) \times \{w \in H^1((0, 1)) : w(x) \geq 0 \text{ a.e. } x \in (0, 1)\}$
Dual Cone	K^*	$\{y \in H^1(0, 1) : \langle x, y \rangle \leq 0, \forall x \in K_0^+\}$
Eigenvalue, eigenfunction	(λ, w^*)	
First critical load	$\bar{\epsilon}_1^c$	
Critical loads	$\bar{\epsilon}_*^c$	
n -th critical load	$\bar{\epsilon}_n^c$	
FEM Residual vectors	$\mathbf{R}, \tilde{\mathbf{R}}$	
FEM base functions and derivatives	$\mathcal{N}_i, \mathcal{N}'_i$	
Eigenvalue at load t	λ_t	
FEM state, generic	$\mathbf{X}$	
FEM new state	$\mathbf{X}'$	
FEM current equilibrium state	$\mathbf{X}^*$	
FEM perturbation	$\mathbf{p}$	

Table 1: Notation and Definitions. The subscripted t in $H_t^1(0, 1)$ indicates a t -parametrised boundary datum

{table:notation}

accessible during loading spawn many possible evolution paths. One can expect that under a quasi-static loading protocol, the system navigates among metastable states which are continuous branches of equilibria. These branches can bifurcate and intersect as well as terminate at points where the state stability is lost. At an instability threshold, the system restabilizes in a dissipative manner through a state transition, whether smoothly or suddenly. In this quasi-static setting, how does the system choose a new locally stable equilibrium branch with lower energy? During an isolated switching event, new equilibrium branches can be determined using a steepest descent or a continuation algorithm. The path selection may suffer from indeterminacy, however, because the energy functionals in our focus are strongly nonlinear, they lack convexity in their argument and thus exhibit multiple local minima, or none at all. Consequently, conducting stability and bifurcation analyses becomes crucial to distinguish among the various potential solutions or evolution paths, those that are physically relevant.

Bifurcation and stability of equilibrium configurations in dynamic systems without constraints has led to a systematic investigation of local blow-up behaviors at bifurcations points in terms of linearised

(canonical) representations, allowing for easier classification and analysis of the bifurcation types [38]. For systems of ODEs the criterion of bifurcation (from a fundamental solution) amounts to the study of the existence of solutions different than the fundamental one in an arbitrary neighborhood of the control parameters. Conditions of failure of the implicit function theorem [38] describe the scenario under which a system of equations can realize more than one smooth solution. Less clear is the picture in presence of nonconvexities and nonlinear constraints associated with internal variables, where quasi-static evolutionary problems defined by optimality conditions take the form of variational inequalities defining the trajectories of a system in phase space. In these scenarios, as noted in the seminal work [34], the study of bifurcation and stability is not equivalent to the existence of solutions infinitesimally near critical points in arbitrary neighborhoods of the control parameters [8].

In our context, a bifurcation condition along the system’s evolution parametrized by the control parameter(s) is associated with the uniqueness of a field of vectors tangent to the trajectory in phase space. The fracture of brittle thin films bonded to substrates unveils a myriad of complex crack patterns, as evidenced by numerical studies [6, 2, 35, 67, 7], resembling those observed in natural contexts such as sand or dried mud [30], and even in biological structures like animal skin [63] and bark [17, 72]. These observations hold particular relevance in the domain of stretchable and flexible electronics [26, 29] including self-healing metal thin films on a flexible substrates [79].^{ALB} In this work, we aim at characterising the stability (or observability) of static solutions (at a given control parameter) as well as to describe the evolutionary paths stemming from an initial condition. Conditions for uniqueness of the evolution path (or its non-bifurcation) reduce to the uniqueness of solutions to a boundary value problem defined for the *rates of evolution*, or equivalently, the positive definiteness of its bilinear operator in a vector space.

Stability is a conceptually different notion when constraints play a role. The loss of such a property for a stationary points of an energy functional is of paramount importance in materials science and engineering. Illustrative in this sense are Euler buckling [10], wrinkling in thin films [37], homogeneous nucleation of dislocations in a crystal [16, 61, 3, 52, 5], buckling of lattice structures [20, 9], nucleation of cracks in soft solids or in pantographic structures [64, 67], plastic avalanches in crystals or amorphous materials [89, 83, 88].

The absence of analytical solutions in strongly non-linear settings requires resorting to numerical methods for computing and predicting equilibrium configurations that correspond to the minima of an energy functional. The minimization process involves discretizing the continuum fields onto a computational grid using methods such as finite elements, finite differences, or spectral techniques [66, 47, 36]. Afterwards, an iterative solver is employed to seek equilibrium energy states, with options including the Newton-Raphson method [84], fixed-point iteration [18, 40, 75], line-search-based descent algorithms like steepest descent or conjugate gradient [74, 23], quasi-Newton methods such as the highly-efficient Limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) approach [46] which involves approximating the Hessian matrix, or more recent advancements like the fast inertial relaxation engine (FIRE) [32]. Recently, several authors [11, 57, 85] demonstrated the efficiency of the quasi-Newton BFGS method for the fully coupled phase-field fracture problem. In particular, [57] reported reduction of computation times by several orders of magnitude with the BFGS method, and the number of load iterations required by a staggered solution scheme as 3000 times higher than the number of iterations required by the BFGS method.^{OUS} These solvers iteratively refine solutions starting from an initial guess provided as part of the solution procedure. Despite their widespread application, there remains a lack of clear understanding regarding the performance of these algorithms and their effectiveness in locating local minima.

We build upon the the work of [7] providing a proof of concept for the numerical computation of stable solutions of phase-field fracture evolutionary models, extending the analysis beyond the second order bifurcation problem and addressing the full nonlinear stability of the solutions. Our investigation starts from the unconstrained case where damage and cracks can heal (without energy expense), which is unrealistic in the applications but allows to thoroughly investigate the system’s bifurcations and to highlight the main mathematical differences between the reversible and irreversible cases. To this end,^{ALB} we consider two one-dimensional phase-field fracture models of a brittle membrane on two types of substrates: one stiff, one compliant. The first model describes a brittle thin film deposited on a stiff (rigid) substrate, while the second model involves a compliant yet unbreakable substrate that can undergo non-uniform deformations. The finite stiffness of the substrate in the second scenario leads to nontrivial qualitative differences in terms of uniqueness of the evolution path, associated with the loss of stability of the unfractured solution [6, 42, 7, 33, 90].

Despite the one-dimensional setting we adopt here which allows for analytical predictions, these mod-

els reveal a complex landscape of equilibrium states with multiple local minima. In the absence of an irreversibility constraint, bifurcation points from homogeneous solution can easily be calculated analytically and numerically, by employing continuation techniques. An *equilibrium map* can be constructed in this setup, allowing all inhomogeneous solutions connected to the homogeneous branch to be identified along with their stability. This enables us to monitor the solutions returned by various numerical optimization techniques and assess their observability. Our findings indicate that under quasi-static loading conditions, line-search-based descent algorithms not relying on full Hessian can fail to detect expected branch-switching events and may return solutions that persist on unstable branches, thus lacking physical relevance. We propose a remedy to this situation which involves utilizing information from the Hessian of the functional when it becomes singular. To discuss this scenario we distinguish two settings, namely i) that in which damage is reversible and all small perturbations are admissible, and ii) the case where damage is subject to an irreversibility constraint which forbids healing. In the former scenario negative variations of damage are allowed and indeed may occur - if convenient from an energetic viewpoint. In the second setting, instead, we consider damage as a unilateral irreversible process stemming from an irreducible one-directional pointwise growth constraint. We emphasize that we consider the reversible case as a prototypical study, rather than for its general physical relevance. Not only because it allows us to construct an *equilibrium map*, but also because it allows to highlight on physical grounds the mathematical differences **in the second-order notions of bifurcation and stability**^{ALB} between the reversible and irreversible cases. The study of the reversible setup may still be relevant in certain phase-field damage models where irreversibility is imposed only on crack sets that exceed a *given* damage threshold, referred to as relaxed crack-set irreversibility [15, 43, 24], or in models with softening elastic energy without irreversibility constraint [80, 68, 67, 4].

The rest of the paper is organized as follows. In Section 2, we present one-dimensional phase-field fracture models with both rigid and compliant elastic foundations. In Section 3 we focus on the analysis of linear and non-linear stability regarding trivial solutions. In Section 4, we construct the equilibrium map in the reversible setup, discuss the selection of equilibrium branches using various numerical optimization algorithms, and explore how irreversibility affects the stability of solutions. In the final Section 5 we summarize our results.

Notation. We employ standard notation for scalar Sobolev spaces defined on the unit interval, such as $H^1(0, 1)$, derivatives of one-dimensional fields, and matrix indices. We indicate the $L^2(0, 1)$ -inner product of functions u, v by $\langle u, v \rangle = \int_0^1 u v dx$. Subscripted t means t -parametrised quantities, superscripted (k) means k -th iterate of an iterative algorithm. We indicate with boldface letters finite element matrices and vectors. We use the prime sign to indicate spatial derivatives. Throughout the investigation we consider a bilayer structure. We use American English spelling throughout the text.

2 Material, Structure, and Evolution

Two one-dimensional fracture models of brittle thin films bonded to substrates with different mechanical properties provide a framework to investigate the evolution and stability of crack patterns under external loading, in a simple scenario where multiplicity of solutions, equilibrium bifurcations, and stability transitions interplay.

Material Model We consider a one-dimensional isotropic and homogeneous brittle material modelled by a state function $W(e, \alpha, \alpha')$ which, at any point x , depends on the local membrane strain $e(x)$ (associated with in-plane displacements $u(x)$, namely $e(x) = u'(x)$), the local damage $\alpha(x)$, and the local gradient of the damage $\alpha'(x)$. Here, the damage variable α is a scalar field driving material softening, bounded between 0 and 1, where 0 indicates the undamaged material and 1, the cracked material. Thus, at points where $\alpha = 0$ the material is elastic with a stiffness E (its Young modulus), at points where $\alpha = 1$ the material has a crack and zero residual stiffness, whereas for intermediate damage values the material's stiffness is $0 < E\mathbf{a}(\alpha) < E$. The state function W is defined as

$$W(e, \alpha, \alpha') := \frac{1}{2}E\mathbf{a}(\alpha)e^2 + \mathbf{w}(\alpha) + \frac{\ell^2}{2}\alpha'^2, \quad (1) \quad \{\text{def:energy_m}$$

where $\mathbf{w}(\alpha)$ can be interpreted as the energy dissipated during an homogeneous damaging process. It is combined with a term proportional to the square of its gradient which controls the energy cost of spatial damage variations. In the first summand, $\mathbf{a}(\alpha)$ is the function that describes the material softening. For

physical consistency, $\mathbf{a}(\alpha)$ is a non-negative function monotonically decreasing from 1 as α increases, reaching zero for $\alpha = 1$. On the other hand, $\mathbf{w}(\alpha)$ is a non-negative, zero only if $\alpha = 0$, and monotonically increasing with α , reaching $\mathbf{w}(1) = 1$. The damage-dependent stress is $\sigma(\alpha) := E\mathbf{a}(\alpha)e$. The parameter ℓ is a characteristic length that controls the competition between localisation and homogeneous damage, effectively controlling the width of damage localisations, the peak stress of the material in one-dimensional traction experiments, and - more in general - structural size effects. Specifically, both functions $\mathbf{a}(\alpha)$ and $\mathbf{w}(\alpha)$ are chosen to be quadratic, namely

$$\mathbf{a}(\alpha) = (1 - \alpha)^2, \quad \mathbf{w}(\alpha) = \mathbf{w}_1 \alpha^2, \quad (2) \quad \{\text{def:constitu}\}$$

This modelling choice is common (yet not unique) in phase-field fracture models (cf. [15, 53, 54]). In the current context, it allows damage to evolve for an arbitrarily small value of the load. The choice of a quadratic damage potential introduces the fundamental conceptual (and computational) difficulty of the evolution law already at the initial loading step. In contrast, the commonly used model with $w(\alpha)$ linear in α (usually referred to as AT1) features an elastic regime in which damage variations are ruled out by optimality. Consequently, second order conditions (bifurcation and stability) are trivial, because damage cannot evolve. In our quadratic model, however, damage can evolve from the very first discrete load increment, making the space of admissible damage variations nontrivial at all positive load levels. Specifically, in the irreversible scenario, admissible perturbations constitute a *solid* cone, requiring careful handling of the inherent nonlinearity at both first and second orders. Thus, our model choice highlights a significant numerical challenge arising immediately at the initial time step, clearly distinguishing reversible from irreversible stability and bifurcation problems. Furthermore, this model naturally enforces the physical bounds on the damage variable (0 and 1) through energy minimality alone. Hence, no additional numerical constraints are required at first order, greatly simplifying numerical implementations and enabling exploration of the complex bifurcation landscape using classical continuation methods. ^{ALB}

Structural Model The structure under consideration is a multilayer composite constituted by a brittle thin film made of the material identified by the state function W , attached to an underlying substrate which is either ^{stiff}rigid^{ALB} or elastically compliant. The thin film is a one-dimensional membrane with thickness h and length L with $L \gg h$, subject to a combination of ~~imposed inelastic tensile strains~~ $\bar{\epsilon}_t^{\text{ALB}}$ imposed displacements by the substrate and loadings at the boundary. The structure's reference configuration is the interval $(0, L)$. The substrate is modelled as a one-dimensional elastic foundation whose displacement field is $v(x)$ ~~and strain field is $\epsilon(v) = v'(x)$~~ ^{ALB}. The displacement field which is elastically compatible to an homogeneous strain in the substrate is the linear function $v(x, t) = \bar{\epsilon}_t/2(2x - 1)$, where $\bar{\epsilon}_t \in \mathbb{R}^{\text{ALB}}$ is the ~~applied tensile~~^{ALB} ~~average~~^{ALB} strain and $t > 0$ plays the role of a loading time parameter. The film is subjected to the displacement of the substrate^{ALB} $v(x, t)$ ~~as a tensile imposed load~~^{ALB} and to given (compatible) displacements at its free ends $x \in \{0, L\}$, so that for all t , $u(0) = v(0, t)$ and $u(L) = v(L, t)$. Our first model describes a brittle thin film deposited on a stiff, non-deformable substrate. This model assumes the substrate is ^{stiff}rigid^{ALB}, meaning, v is a given. The elastic interaction is modelled by a distributed linear elastic foundation of stiffness K , thus the total energy of the structure is a functional Ψ constructed by considering the energy of the thin film and the energy associated with the mechanical coupling between the film and the substrate. In nondimensional form, it is given by

$$\Psi(u, \alpha) = \int_0^1 \left[\frac{1}{2} \mathbf{a}(\alpha)(u')^2 + \frac{1}{2\Lambda^2}(u - v)^2 + \mathbf{w}(\alpha) + \frac{\ell^2}{2}(\alpha')^2 \right] dx, \quad (3) \quad \{\text{def:energy}_s\}$$

where $\Lambda^2 = \frac{E_{\text{eff}}}{K}$ ^{ALB}, E_{eff} being the effective stiffness of the ~~onetwo~~^{ALB} dimensional ~~filmmembrane~~^{ALB} and spatial variables as well as physical displacements are respectively normalized with respect to the film's length L and the displacement scale $u_0 := \frac{w_1 L}{K}$ ^{ALB}. Our second model involves the same brittle thin film but a *compliant* elastic substrate that undergoes deformation alongside the film. Unlike for the rigid substrate, the substrate's deformation is an additional unknown which affects the overall energy landscape of the system, incorporating an extra term accounting for the strain energy of the substrate. The state of this structure is identified by the triplet $y := (u, \alpha, v)$, and the nondimensional energy of the compliant system $\tilde{\Psi}(u, \alpha, v)$ reads

$$\tilde{\Psi}(u, \alpha, v) = \int_0^1 \left[\frac{1}{2} \mathbf{a}(\alpha)(u')^2 + \mathbf{w}(\alpha) + \frac{\ell^2}{2}(\alpha')^2 + \frac{1}{2\Lambda^2}(u - v)^2 + \frac{\rho}{2}(v')^2 \right] dx. \quad (4) \quad \{\text{def:energy}_c\}$$

[ALB 1] double check denominator

The reduced elastic model in (3) (and (4)) can be obtained as the asymptotic vanishing thickness limit for a thin 3d elastic bilayer system of thickness $h = o(L)$, as shown in [44]. Assuming a three-dimensional system where the film is attached to its substrate by the means of a thin interface layer of thickness $h' = O(h)$ and of stiffness $E'_{2d} \ll E_{2d}$, the nondimensional constants appearing in the expression of the reduced-dimension energies can be related to the three-dimensional parameters of a physical mechanical system.^{ALB} Denoting by E_{2d} and h the two-dimensional stiffness (per unit depth) and the thickness of the film membrane, and using a prime sign to indicate the stiffness and thickness of the substrate, the nondimensional quantities appearing in the expression of the energy depend on the geometric the elastic properties of the 3d system as follows

$$\ell := \frac{\bar{\ell}}{L}, \quad \Lambda := \frac{E_{2d}}{E'_{2d}} \frac{hh'}{L^2}^{\text{ALB}}, \quad \rho := \frac{E'_{2d}}{E_{2d}} \frac{h'}{h}^{\text{ALB}}, \quad (5) \quad \{\text{def:dimension}\}$$

Note that the nondimensional parameter Λ relates the effective stiffness of the film membrane undergoing planar tensile strains (i.e. $E_{2d}h$), to the effective stiffness of the bonding layer which, to accommodate the mismatch deformation between the film and the underlying substrate, is subject to a simple shear for which the effective stiffness is inversely proportional to its thickness, namely E'_{2d}/h' .^{ALB}

For brevity, we denote by $y := (u, \alpha)$ (and $y := (u, \alpha, v)$) the mechanical state of the stiff (respectively, compliant) model system, and by $H^1(0, 1)$ the (Sobolev) space of scalar real functions defined on the unit interval which are square integrable and have square integrable first derivatives. The energy $\Psi(y)$ is well-defined for pairs belonging in the Cartesian vector product space $V := H^1(0, 1) \times H^1(0, 1)$. Similarly, the energy $\tilde{\Psi}(y)$ is well-defined for triplets (u, α, v) in $\tilde{V} := V \times H^1(0, 1)$.

Evolutionary model Assuming small enough loading rates, the evolution problem of the structure can be cast in an energetic variational formulation as an incremental rate-independent quasi-static process driven by an energy-minimality principle. This allows identifying sequences of equilibrium configurations as critical states, to ascertain their stability, and determining the system's transition trajectories between different equilibrium states. Optimality conditions defining this problem are derived from the intuitive idea that a state is observable only if it is stable, and in turn, a state is stable only if it is a local minimum of the energy among admissible state perturbations, at a given load level.

Differently from dissipative evolutions driven by an energy gradient flow whereby a system *reaches* equilibrium conditions through a gradient descent process parametrized by an internal timescale, the quasi-static evolution we consider is a sequence of *attained* equilibrium states as a necessary condition for local energy minimality. The rate-indipendancy further implies that the system does not exhibit internal timescales. As such, energy minimisation is performed at any given value of the load, and at each increment of the external load its configuration evolves subject to imposed boundary conditions and possible internal constraints.

Specifically, we consider an evolution during the loading interval $t \in [0, T]$ as a time-parametrized mapping $t \mapsto y_t$ such that, for all $t \in [0, T]$ the realised (observed) state of the system y_t is a local energy minimum among all admissible state perturbations, or with respect to all admissible competitor states. In practice, given an initial condition y_0 at $t = 0$ we seek a state $y_t := (u, \alpha)_t$ (and $\tilde{y}_t := (u, \alpha, v)_t$ respectively, for the compliant model) such that, for a given value of the control parameter t , it satisfies time-dependent kinematic boundary conditions on the displacement variable and is locally energy minimal. For definiteness, denoting by $X_t = \{v \in H^1(0, 1) : v = v(x, t), \text{ for } x = 0 \text{ and } x = 1\} \times H^1(0, 1)$ the affine vector space of (kinematically) admissible states for the stiff model, (respectively $\tilde{X}_t = X_t \times H^1(0, 1)$, for the compliant model) we seek **stable states exploiting a notion of directional stability**.^{ALB} The system at y is directionally stable in the admissible direction z if:^{ALB}

$$\exists \bar{h} > 0 \text{ such that } \Psi(y) \leq \Psi(y + hz), \quad \forall h \in [0, \bar{h}], \quad (6) \quad \{\text{eq:variation}\}$$

^{ALB} (and similarly for the compliant model substituting Ψ with $\tilde{\Psi}$ and X_t with $\tilde{X}_t$.) Here z is any admissible perturbation direction in the tangent space $T_y X_t$. If this condition is satisfied for all admissible directions $z \in T_y X_t$, the state y is considered stable. Remark that this statement focuses on *local* variations of the energy Ψ around y , in line with numerical methods that explore the solution landscape directionally (e.g., quasi-Newton or Newton methods, even in finite-dimensional approximations), yet is defined through a *global* quantity, the energy of the system Ψ . For small perturbations, this is a practical criterion that can be tested numerically without requiring a global search which is computationally prohibitive in infinite-dimensional spaces due to the vastness of the energetic landscape. In physical systems, stability often depends on small perturbations to the current state. Accordingly, this statement

ensures that no small admissible perturbation z decreases the energy of a stable state, which is a natural generalisation of a second-order stability condition. This condition is less restrictive than global energy minimisation but stricter than mere stationarity and is applicable for both reversible and irreversible systems since the admissibility of directions z can incorporate irreversibility constraints, ensuring that stability is tested only along physically meaningful directions. Compared to global minimisation, this approach is less restrictive, more realistic, in line with numerical methods, and provides a rigorous stability framework with a clear and physically meaningful *global* condition for (local, directional) stability. In infinite-dimensional spaces, the notion of locality depends on the choice of topology (norm), which undermines the physical meaning of the solution, as the system's behavior should not depend on an arbitrary mathematical choice. The directional minimality statement above is appealing because it avoids the pitfalls of both global and norm-dependent local minimality, by leveraging the topology of the real line which is intrinsic. This aligns with physical intuition that systems tend to evolve along a *path* according to stability and constraints. ^{ALB}

Admissibility of competitors and perturbations The admissibility of state competitors explicitly depends on the loading parameter through the kinematic boundary conditions (on displacement) and on internal constraints, namely whether damage (and hence the softening material behaviour) evolves in a reversible or irreversible manner. In the first case, as damage can evolve freely within the interval $[0, 1]$, all admissible states in X_t (respectively, in $\widetilde{X}_t$) are also admissible competitors. In the second case, the damage field is subject to the pointwise irreversibility constraint $\dot{\alpha}(x) \geq 0, \forall x \in [0, 1]$, requiring that the damage can only increase or stay constant. As a consequence, irreversibility restricts the admissible set of competitors to the set $K_{\alpha_t}^+ := H^1(0, 1) \times \{\beta \in H^1(0, 1) : \beta \geq \alpha_t\}$ (respectively, $\widetilde{K}_{\alpha_t}^+ := K_{\alpha_t}^+ \times H^1(0, 1)$). Remark that in the definition of the competitor space α_t (the damage field at time t) is unknown at time t . In the irreversible case the set of admissible competitor states depends explicitly on the entire history of the evolution through the current damage field. To draw the attention to the consequences of irreversibility on the system's transitions between different equilibrium states we develop the global variational inequality (6) (and the analogous for the compliant model) by expanding the energy around the state y_t . Admissible perturbations in the fully reversible case belong to the *tangent*^{ALB} space $T_{y_t}X_t = X_0^{\text{ALB}}$ associated with X_t (respectively, $\widetilde{X}_t$) which is a linear vector space^{ALB}, whereas in the irreversible case *admissible perturbations constitute they are bound to*^{ALB} the closed, *pointed*^{ALB}, convex cone K_0^+ (respectively, $\widetilde{K}_0^+$).

An energy expansion in the vicinity the state y_t reads

$$\Psi(y) - \Psi(y_t) = \delta\Psi(y_t)(y - y_t) + \frac{1}{2}(y - y_t)^T \delta^2\Psi(y_t)(y - y_t) + o(\|y - y_t\|^2),$$

(and analogously for the compliant model), which allows to write first and second order (necessary and sufficient) conditions for optimality.

In the subsequent section, we exploit the energy-stability inequality (6) ~~for both energy expansion models~~ introduced above, exploring the equilibrium configurations and identifying the conditions under which the system transitions between different equilibrium branches.

3 Linear and nonlinear stability of homogeneous solutions^{ALB}

Solutions to the incremental evolutionary problem are sought by solving first and second order necessary conditions for optimality encoded in the global variational inequality (6) which reduces to a *linear* stability problem in a linear vector space for the case of fully reversible damage, and constitutes an instance of a *nonlinear* stability problem in a convex cone in the presence of irreversibility. {sec:stability}

Linear stability in the reversible case - stiff substrate Equilibrium equations of the system are obtained as first order necessary conditions for energy minimality, satisfying imposed displacements at the ends of the film. The first order variation of the energy functional Ψ in the direction $z := (w, \beta)$ is given by the following linear form

$$\delta\Psi(u, \alpha)(w, \beta) = \int_0^1 \left[\mathbf{a}(\alpha)u'w' + \frac{1}{\Lambda^2}(u - v)w + \left(\frac{1}{2}u'^2\mathbf{a}'(\alpha) + \mathbf{w}'(\alpha) \right) \beta + \ell^2\alpha'\beta' \right] dx, \quad (7) \quad \{\text{firstvar1}\}$$

where $z \in X_0$ is a test function (an admissible perturbation) in the linear space associated with X_t . First order minimality conditions are *local* conditions associated with the stationarity of the energy functional.

The first order variation should vanish for all admissible test functions, namely

$$\delta\Psi(u, \alpha)(w, \beta) = 0, \quad \forall z \in X_0. \quad (8) \quad \{\text{eq:stationar}\}$$

By using standard arguments of the calculus of variations, localizing the integral and choosing $\beta = 0$ first, and then $v = 0$ leads to establishing the strong form of local stationarity conditions. The mechanical equilibrium, the damage criterion, and the associated boundary conditions, ^{ALB} are thus respectively given by

$$\begin{cases} 2(1-\alpha)\alpha'u' + (1-\alpha)^2u'' - \frac{1}{\Lambda^2}(u-v) = 0, & x \in (0, 1) \\ -\ell^2\alpha'' - (1-\alpha)(u')^2 + 2\alpha = 0, & x \in (0, 1) \end{cases}, \quad \begin{cases} u(x) = v(x, t), & x = \{0, 1\}, \forall t \text{ ^{ALB} } \\ \alpha'(x) = 0, & x = \{0, 1\}, \forall t \end{cases}. \quad (9) \quad \{\text{auto1}\}$$

Notice that the choice of boundary conditions for displacements compatible with the substrate's deformation implies that the pair $y^{\text{hom}}(\bar{\epsilon}) := (u_h(x), \alpha_h)(\bar{\epsilon})$ given by $u_h(x) \equiv v(x, t)$ and α_h a load-dependent constant to identify, is always (the unique homogenous) solution to the first order equilibrium equations. This makes it immediate to identify the fundamental homogeneous solution branch $t \mapsto y^{\text{hom}}$ and to decouple the elasticity problem from the evolution of damage. Therefore, the solution to (9) such that $u''(x) = \alpha'(x) = 0, \forall x \in (0, 1)$ is the homogeneous branch

$$u_h(x) = \frac{\bar{\epsilon}_t}{2}(2x-1), \quad \alpha_h = \frac{\bar{\epsilon}_t^2}{2 + \bar{\epsilon}_t^2}. \quad (10) \quad \{\text{eq:homo1}\}$$

Notice that, first order conditions identify the critical load threshold that activates the damaging process. The quantity $\bar{\epsilon}_*^c$, the value of the load for which the undamaged elastic state becomes unstable, is determined injecting the elastic solution u_h in the energy and computing $\delta\Psi(u_h, 0)(0, \beta) = 0$. According to our mechanical energy model, the critical load is given by

$$\bar{\epsilon}_*^c = \sqrt{2 \frac{w'(0)}{a'(0)}}, \quad (11)$$

which, by our choice of material model (whereby $w(\alpha)$ is quadratic, cf (2)) we get that $\bar{\epsilon}_*^c = 0$ so that damage necessarily starts as soon as $t > 0$. The effective total energy along the homogeneous branch reads

$$\Psi^{\text{hom}}(\bar{\epsilon}_t) := \Psi(y^{\text{hom}}(\bar{\epsilon}_t)) = \frac{\bar{\epsilon}_t^2}{2 + \bar{\epsilon}_t^2}. \quad (12) \quad \{\text{eq:energy-ho}\}$$

An equilibrium configuration $y_t := (u, \alpha)_t$ is a state such that the first variation $\delta\Psi(y_t)(z)$ vanishes for all admissible test fields in the vector space X_0 . To assess the incremental stability of the homogeneous solution in the reversible (linear) case, we examine the positivity of the second variation, requiring

$$\delta^2\Psi(y_t)(y - y_t, y - y_t) > 0, \quad \forall y - y_t \text{ ^{ALB} } \in X_0, \quad (13) \quad \{\text{eqn:linear-s}\}$$

The second directional derivative of the energy is given by the following bilinear form

$$\delta^2\Psi(u, \alpha)(v, \beta) = \int_0^1 \left[(1-\alpha)^2 v'^2 + \frac{1}{\Lambda^2} v^2 \right] dx + \int_0^1 \left[-4(1-\alpha)u'v'\beta + (2+u'^2)v^2 + \ell^2\beta'^2 \right] dx, \quad (14) \quad \{\text{hessian22}\}$$

which is well defined for perturbations $(v, \beta) \in X_0$. We extract information on the onset of instability seeking a solution in Fourier series of the fields v and β in (14), such that

$$v(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x + \phi_n), \quad \beta(x) = \sum_{n=1}^{\infty} b_n \cos(n\pi x + \psi_n).$$

Then we observe that, thanks to boundary conditions, $\psi_n = \phi_n = 0$ for all natural n . The stability condition (13) for homogeneous states y^{homALB} takes the form:

$$\begin{bmatrix} a_n & b_n \end{bmatrix} \mathcal{H} \begin{bmatrix} a_n \\ b_n \end{bmatrix} = \begin{bmatrix} a_n & b_n \end{bmatrix} \begin{pmatrix} a(\alpha)(n\pi)^2 + \Lambda^{-2} & \frac{\partial a}{\partial \alpha} \bar{\epsilon}(n\pi) \\ \frac{\partial a}{\partial \alpha} \bar{\epsilon}(n\pi) & \frac{\partial^2 a}{\partial \alpha^2} \bar{\epsilon}^2 + \frac{\partial h^2}{\partial \alpha^2} + \ell^2(n\pi)^2 \end{pmatrix} \begin{bmatrix} a_n \\ b_n \end{bmatrix} > 0. \quad (15) \quad \{\text{hessian1}\}$$

By substituting the homogeneous solution (u_h, α_h) into (15), we compute $\det \mathcal{H}$ as a function of $\bar{\epsilon}$ and n , which is depicted in Figure 1. The calculations are performed for the parameter values $\ell = 0.16$

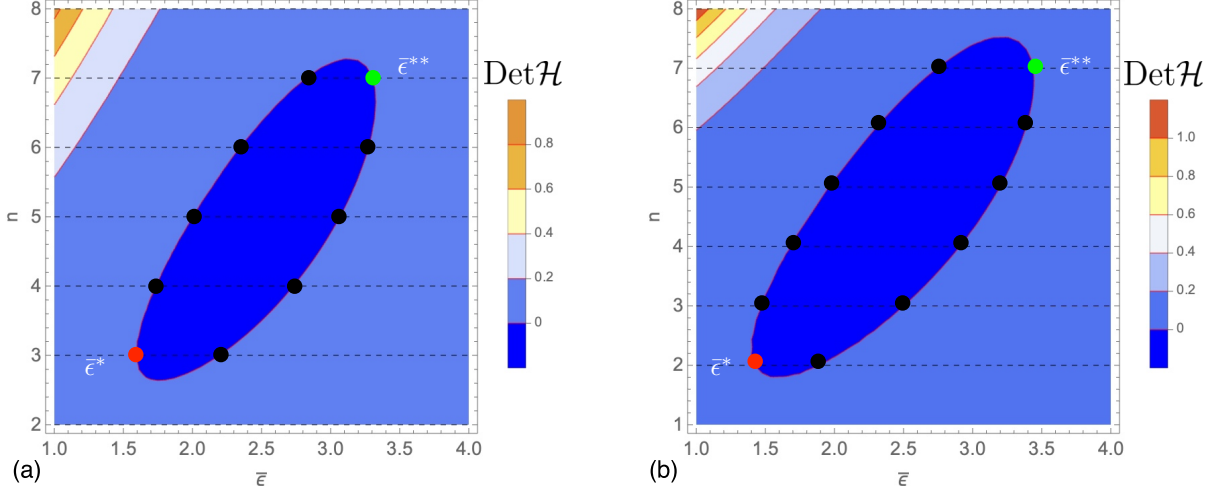


Figure 1: Determinant $\det H$ for homogeneous solution α_h ($\ell = 0.16$, $\Lambda = 0.34$) for: (a) rigid foundation; (b) compliant foundation ($\rho = 0.5$). Closed loops indicate re-entry behavior at large deformations. Black dots: bifurcations from homogeneous solution; red/green dots: critical strains $\bar{\epsilon}^*$ and $\bar{\epsilon}^{**}$ (lower/upper stability limits). Critical wave numbers differ between limits: $n_c(\bar{\epsilon}^*) \neq n_c(\bar{\epsilon}^{**})$.^{OUS}

{fig:hessian1}

and $\Lambda = 0.34$. The figure represents, for a given load $\bar{\epsilon}$, the wave number $n(\bar{\epsilon}) \in \mathbb{N}$ of possible energy-decreasing damage bifurcations. For an increasing loading history $\bar{\epsilon}_t \nearrow$, the wave number is non-monotonic. In Fig. 1(a), the locus $\det \mathcal{H} = 0$ forms closed loops, indicative of an elastic background's influence. Notably, a re-entry behavior of the affine configuration (i.e., the homogeneous configuration re-stabilizes at large deformation) is discernible, marked by the emergence of two critical strains denoted as $\bar{\epsilon}^*$ and $\bar{\epsilon}^{**}$ (with $\bar{\epsilon}^* < \bar{\epsilon}^{**}$), representing the lower and upper stability limits for the homogeneous state. These critical points are highlighted by red and green dots in Fig. 1(a). The critical wave number n_c for the lower limit $n_c(\bar{\epsilon}^*)$ differs from the critical wave number for upper limit $n_c(\bar{\epsilon}^{**})$. Finally, we remark that closed-form analytical solutions can be provided for the critical wave number and critical strains and the parametric dependence of the corresponding bifurcation thresholds can be obtained, as detailed in [67].

Linear stability in the reversible case - compliant substrate Denoting by $z := (w, \beta, \tilde{v}) \in \tilde{X}_0$ a test function for the state triplet $y_t := (u, \alpha, v)_t$ at time t , the first order variation of the energy functional $\tilde{\Psi}$ is given by the following linear form

$$\delta \tilde{\Psi}(u, \alpha, v)(w, \beta, \tilde{v}) = \int_0^1 [(1-\alpha)^2 u' v' + \frac{1}{\Lambda^2} (u-v)(v-\tilde{v}) + \rho v' \tilde{v}' + \frac{1}{2} u'^2 (a'(\alpha) + w'(\alpha)) \beta + \ell^2 \alpha' \beta'] dx, \quad (16) \quad \{\text{firstvar}\}$$

Arguing similarly to the stiff case by localization and integration by parts, for all $x \in (0, 1)^{\text{ALB}}$ the Euler-Lagrange equations and associated boundary conditions read

$$\begin{cases} 2(1-\alpha)\alpha' u' + (1-\alpha)^2 u'' - \frac{1}{\Lambda^2} (u-v) &= 0, \\ -\ell^2 \alpha'' - (1-\alpha)(u')^2 + 2\alpha &= 0, \\ \rho v'' + \frac{1}{\Lambda^2} (u-v) &= 0, \end{cases} \quad \begin{cases} u(x) = v(x, t), & x \in \{0, 1\}, \forall t \text{ ALB} \\ \alpha'(x) = 0, & x \in \{0, 1\}, \forall t \end{cases}. \quad (17) \quad \{\text{auto2}\}$$

It is easy to show that the homogeneous solution on the trivial branch remains the same as in the case of the rigid substrate, that is $\alpha_h(\bar{\epsilon}) = \frac{\bar{\epsilon}^2}{2+\bar{\epsilon}^2}$, whereas the effective elastic energy along the homogeneous branch now reads $\tilde{\Psi}^{\text{hom}}(\bar{\epsilon}) = \frac{\bar{\epsilon}^2}{2+\bar{\epsilon}^2} + \frac{\rho}{2} \bar{\epsilon}^2$.

We once more seek the linear incremental stability of an equilibrium configuration $y_t := (u, \alpha, v)_t$ satisfying that the first order condition $\delta \tilde{\Psi}(y_t)(y - y_t) = 0$ for all admissible state perturbations $z := y - y_t \in \tilde{X}_0$ by examining the positivity of the second variation, namely

$$\delta^2 \tilde{\Psi}(y_t)(y - y_t, y - y_t) > 0, \quad \forall y - y_t \in \tilde{X}_0,$$

The second variation is given by the following bilinear form

$$\delta^2 \tilde{\Psi}(y_t)(y - y_t, y - y_t) = \int_0^1 \left[(1 - \alpha_t)^2 w'^2 - 4(1 - \alpha_t) u'_t w' \beta + (2 + u_t'^2) \beta^2 + \ell^2 \beta'^2 + \frac{1}{\Lambda^2} (w^2 + \tilde{v}^2) + \rho \tilde{v}'^2 \right] dx. \quad (18) \quad \{\text{hessian222}\}$$

We again proceed to extract information on the onset of instability expanding in Fourier series the fields w , β and $\tilde{v}$ appearing in (18) such that

$$w(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x + \phi_n), \quad \beta(x) = \sum_{n=1}^{\infty} b_n \cos(n\pi x + \psi_n), \quad \tilde{v}(x) = \sum_{n=1}^{\infty} c_n \sin(n\pi x + \theta_n).$$

Similar to the previous section we claim that, whenever the first order term vanishes, the system is stable only if $\delta^2 \tilde{\Psi} > 0$ for all sufficiently smooth admissible test fields $(w, \beta, \tilde{v})$ in the vector space $\tilde{X}_0$. The stability condition for the homogeneous state y^{homALB} takes the form

$$[a_n \quad b_n \quad c_n] \begin{pmatrix} a(\alpha)(n\pi)^2 + \Lambda^{-2} & \frac{\partial a}{\partial \alpha} \bar{\epsilon}(n\pi) & -\Lambda^{-2} \\ \frac{\partial a}{\partial \alpha} \bar{\epsilon}(n\pi) & \frac{\partial^2 a}{\partial \alpha^2} \bar{\epsilon}^2 + \frac{\partial w^2}{\partial \alpha^2} + \ell^2 (n\pi)^2 & 0 \\ -\Lambda^{-2} & 0 & \Lambda^{-2} + \rho (n\pi)^2 \end{pmatrix} \begin{bmatrix} a_n \\ b_n \\ c_n \end{bmatrix} > 0. \quad (19) \quad \{\text{hessian3}\}$$

By substituting the homogeneous solution $\alpha_h(u_h, \alpha_h)^{\text{ALB}}$ into (19), we compute $\det \mathcal{H}$, as depicted in Fig. 1(b). The calculations are performed for the parameter values $\ell = 0.16$, $\Lambda = 0.34$ and $\rho = 0.5$. In Fig. 1(b), we again observe the formation of closed loops and a re-entry behavior of the affine configuration is discernible. This is marked by the emergence of two critical strains denoted as $\bar{\epsilon}^*$ and $\bar{\epsilon}^{**}$, representing the lower and upper stability limits for the homogeneous state. These critical points are highlighted by red and green dots in Fig. 1(b). The critical wave number n_c for the lower limit $n_c(\bar{\epsilon}^*)$ differs from the critical wave number for the upper limit $n_c(\bar{\epsilon}^{**})$. The overall behavior of the system remains the same while the first critical wave number $n_c = 2$ is now smaller suggesting a different number of cracks will appear during the loading history.

Nonlinear stability with irreversibility constraints Irreversibility is introduced in the variational formulation of the evolution problem as a pointwise inequality constraint. Intuitively, irreversibility plays two distinct roles along an evolution: it acts (i) as a *local* constraint which prevents the damage field at a given location to decrease between two subsequent loads values (both during monotonic and non-monotonic load programs), and (ii) as a global restriction of the space of admissible variations in such a way that *negative* perturbations of the current damage state are no longer allowed. This changes the topological structure of the set of admissible perturbations, from a linear vector space to a convex cone. To enforce irreversibility we consider only non-decreasing damage evolutions that are sufficiently smooth with respect to the loading parameter, and seek maps $t \mapsto y_t = (u_t, \alpha_t)$ such that $\dot{\alpha}_t \geq 0$, satisfying the minimality condition (6). Because the current state can only be compared to those of with equal or higher damage, the space of admissible perturbations is a convex cone strictly contained in the vector space of admissible unconstrained perturbations, namely $K_0^+ \subset X_0$. Indeed, for $(v, \beta) \in K_0^+$, then $-(v, \beta) \in X_0$ but $-(v, \beta) \notin K_0^+$. This restriction has a profound impact on the variational characterization of local minima and bears consequences on both for the first order (equilibrium) conditions and the second order (stability) problem which become *unilateral* conditions. Equilibrium states $y_t = (u_t, \alpha_t)$ of the irreversible system are hence governed by the following (first order) necessary optimality conditions taking the form of a variational inequality

$$\delta \Psi(y_t)(y - y_t) \geq 0, \quad \forall y - y_t \in K_0^+, \quad (20) \quad \{\text{eq:eq_variational}\}$$

which has to hold for all admissible competitor states y such that $v - u_t \in H_0^1(0, 1)$ and $\beta - \alpha_t \in H^1(0, 1)$ with $\beta \geq \alpha_t$. By testing the elasticity problem for fixed damage, and the damage problem for a given displacement field, we obtain

$$\delta \Psi(y_t)(v - u_t, 0) = 0, \quad \delta \Psi(y_t)(0, \beta - \alpha_t) \geq 0, \quad \forall y - y_t \in K_0^+. \quad (21) \quad \{\text{eq:variational}\}$$

The last two relations are, respectively, the weak form of the mechanical equilibrium conditions and the evolution law for the damage field. The former allows to compute the elastic equilibrium displacement u_t which can thus be eliminated from energy, while the latter governs the evolution of the damage field. Upon elimination of the kinematic field, the variational inequality (21).2 takes a particularly expressive

form when written as a complementarity problem. This highlights the mechanical nature of the damage criterion as a threshold law. Namely, the strict convexity of the elastic model for given damage implies that (21).1 has, for given α , a unique time-parametrized solution $u_t(\alpha)$. Substituting in (21).2 and accounting for the irreversibility constraint we are led to seek a map $t \mapsto \alpha_t$ such that

$$\dot{\alpha}_t \geq 0 \quad -\phi_t(\alpha_t) \leq 0 \quad \phi_t(\alpha_t)\dot{\alpha}_t = 0. \quad (22) \quad \{\text{eq:complemen}\}$$

Here, ϕ_t is the scalar function associated with the variation of elastic energy density **computed**^{ALB} at the equilibrium **and**^{ALB} defined by $\delta\Psi(y_t)(0, \beta) = \langle -\phi_t(\alpha_t), \beta \rangle$. Consequently, $\phi_t(0)$ is the variation of the energy density at equilibrium for the undamaged structure, and all equilibrium solutions u_t such that $-\phi_t(0) > 0$ belong to the interior of the damage yield surface for the sound structure. The equality $-\phi_t(0) = 0$, conversely, identifies the elastic limit and indicates that the damage criterion has been attained by the (sound) structure, or equivalently, that the state $(u_t, 0)$ has reached, from the interior, the boundary of the (damage-dependent) elastic domain. Explicitly, the function ϕ_t depends on the brittle material model and is defined, for both the stiff and compliant substrate models, as

$$\phi(\alpha) := \frac{1}{2}\mathbf{a}'(\alpha)\epsilon^2 + \mathbf{w}'(\alpha), \quad (23) \quad \{\text{eq:energy-re}\}$$

where ϵ is the elastic strain and $\mathbf{a}'(\alpha)$ and $\mathbf{w}'(\alpha)$ are the derivatives of the softening and dissipation energy densities with respect to the damage variable. The inequality (21).2 identifies the domains of admissible strains (and, by duality, of stresses) for homogeneous solutions as $\mathcal{R}(\alpha) := \{\epsilon \in \mathbb{R}_{sym}^{2 \times 2} : \mathbb{E}_{2d}\epsilon^2 \leq -\frac{2\mathbf{w}'(\alpha)}{\mathbf{a}'(\alpha)}\}$, and $\mathcal{R}^*(\alpha) := \{\sigma \in \mathbb{R}_{sym}^{2 \times 2} : \frac{\sigma^2}{\mathbb{E}_{2d}} \leq \frac{2\mathbf{w}'(\alpha)}{\mathbf{a}'(\alpha)}\}$, respectively. **As a first order optimality condition, (21).2 states that the local elastic energy release is either smaller than or equal to the (marginal) cost of damage, whereas the complementarity condition $\phi_t(\alpha_t)\dot{\alpha}_t = 0$ ensures that the damage field evolves only if the energy release rate is critical. The three conditions above, established as necessary first order condition for constrained optimality, encode the pointwise non-negativity of the damage rate, the boundedness of the elastic domain, its dependence upon damage, and the complementarity between the attainment of the damage criterion and the conditions for the evolution of the internal order parameter, constitute the set of first order Karush-Kuhn-Tucker conditions the complementarity problem, [?]^{ALB}.**

[ALB 2] non-dimensional? and the yong modulus?

In the current one-dimensional setup with homogeneous initial conditions $y_0 = (0, 0)$ and compatible kinematic boundary conditions, the existence of a homogeneous solution implies that the damage criterion is attained everywhere throughout the bar at the same load. This greatly simplifies the analysis of the energetic properties of the system. Using the elastic solution $u_t = 2\bar{\epsilon}_t(x - 1/2)$ the inequality in (21).2 yields the following algebraic inequality

$$0 \geq -\phi_t(\alpha_t) = -\frac{1}{2}\bar{\epsilon}_t^2 \mathbf{a}'(\alpha_t) - \mathbf{w}'(\alpha_t), \quad (24) \quad \{\text{eq:variation}\}$$

which identifies the evolution of the homogeneous damage response, as a function of the given load level t . The investigation of the stability properties requires considering second order energy variations with respect to all admissible perturbations that render null the first order term in the energy expansion (2). In the general case, this requires distinguishing between the regions where the damage criterion is attained, and thus damage can evolve, from the (complementary) domain where damage cannot evolve (there, the second relation in (22) is satisfied with a strict inequality). In our setup, the existence of nontrivial homogeneous solutions simplifies the analysis because the damage criterion is attained everywhere thus the function space of admissible perturbations is defined on the fixed domain $(0, 1)$.

Assume now that a state y_t is known as a function of t such that it solves (20) and is sufficiently smooth so that the (right) derivative with respect to t is well-defined. As t varies, y_t describes a (smooth) curve in the phase space identified by its right tangent vector $\dot{y}_t =: \lim_{\tau \rightarrow 0^+} \frac{y_{t+\tau} - y_t}{\tau}$, the rate of evolution. A fundamental question is to discern whether y_t is an isolated equilibrium state tracing a unique evolution path, or conversely if it lays at the intersection of multiple equilibrium curves. To this end, differentiating (20) with respect to t we obtain a boundary value problem relating the rate of evolution $\dot{y}_t$ to the current state y_t , supposing the latter known, namely

$$\delta^2\Psi(y_t)(\dot{y}_t, \zeta - \dot{y}_t) + \delta\dot{\Psi}(y_t)(\zeta - \dot{y}_t) \geq 0, \quad \forall \zeta \in X_0, \quad (25) \quad \{\text{eq:variation}\}$$

where $\delta\dot{\Psi}$ is the time-derivative of the linear form corresponding to the first order energy variation. By construction, the homogeneous rate $\dot{y}_h$ is a solution of (25), the question is whether another solution

exists. The uniqueness is thus ensured by the positive definiteness of the quadratic form in X_0 . Thus, the non-bifurcation condition for the homogeneous evolution reads

$$\delta^2\Psi(y^{\text{hom}})(\zeta, \zeta) > 0, \quad \forall \zeta \in X_0, \quad (26) \quad \{\text{eq:bifurcation}\}$$

which formally coincides with the classical *linear stability* problem (13) in the reversible case, yet has a different mechanical interpretation in terms of evolution rates and uniqueness of equilibrium branches. Remark that, in the general case in which the damage criterion is not attained everywhere, the space of admissible perturbations for the (second order) bifurcation problem is $X'_0 := H_0^1(0, 1) \times \{\beta \in H^1(0, 1) : \delta\Psi(y_t)(0, \beta) = 0\}$ for the stiff substrate model, and $\widetilde{X}'_0 := X'_0 \times H_0^1(0, 1)$ for the compliant substrate model. The first load at which the bifurcation inequality (26) fails, namely $t_b := \inf_t \{\delta^2\Psi(y_t)(\zeta, \zeta) = 0, \forall \zeta \in X_0\}$ corresponds to the first bifurcation load, namely, the load for which there exist (multiple) equilibrium curves intersecting the homogeneous branch. As a consequence, for $t \geq t_b$ the possibility exists of bifurcating away from homogeneous branch. The study of the bifurcation problem is functional to infer a partial response on the stability of the state. Indeed, if the current equilibrium branch is unique then, necessarily, the current state is stable. The converse is not true, however, as the existence multiple possible of bifurcation paths is not a sufficient condition to exclude the stability of the current state. This holds true for the irreversible case, due to the conceptual difference between the bifurcation and the stability problems.

The stability of the homogeneous solution in the irreversible case, according to our energetic viewpoint, is governed by the positivity of $\delta^2\Psi(y^{\text{hom}})$ on the constrained space of admissible state perturbations K_0^+ . Denoting t_s the load at which the homogeneous solution loses stability by analogy to the bifurcation load, namely $t_s := \inf_t \{\delta^2\Psi(y^{\text{hom}})(\zeta, \zeta) = 0, \forall \zeta \in K_0^+\}$, the set inclusion $K_0^+ \subset X_0$ implies that necessarily $t_b \leq t_s$. Equality occurs when the first bifurcation mode has nonnegative components.

This indicates a qualitative conceptual distinction between the bifurcation and the stability thresholds, in the irreversible case. As a consequence, a system can persist along a critical non-unique equilibrium branch, yet be stable. A sufficient condition for the stability of the homogeneous state y^{hom} is given by the strict positivity of the Hessian form on the constrained space of admissible perturbations, namely (for the stiff substrate model)

$$\delta^2\Psi(y^{\text{hom}})(y - y^{\text{hom}}, y - y^{\text{hom}}) > 0, \quad \forall y - y^{\text{hom}} \in K_0^+, \quad (27) \quad \{\text{eq:variational}\}$$

and similarly for the compliant substrate model, by replacing Ψ with $\widetilde{\Psi}$ and K_0^+ with $\widetilde{K}_0^+$. The variational inequality above is a constrained eigenvalue problem which we exploit to characterize the stability of the state y^{hom} . Its solution yields, at load ϵ_t , either a positive eigenpair $(\lambda_t, z_t^*) \in \mathbb{R}^+ \times K_0^+$ as a sufficient condition for the stability of current state, or a pair $(\lambda_t, z_t^*) \in \mathbb{R}^- \times K_0^+$ where λ_t is the local (negative) energy curvature and the eigenmode z_t^* , indicating the direction of maximum energy decrease is interpreted as the *instability mode*, pointing the system towards an optimal direction of energy descent. From the numerical standpoint, the bifurcation eigen-problem in the vector space (25) may be regarded as an approximated version of the stability problem, in the time-discrete setting. As suggested in [7] through the notion of ‘incremental-stability’, the irreversibility constraint in the stability problem can be relaxed to a pointwise inequality with respect to the state at the previous time-step, denoted y_- . The mechanical intuition is to replace the *current* state y_t with y_- , in the definition of the space of perturbations. As a consequence, denoting by α_t the equilibrium damage field solving first order optimality conditions at t and by α_- is the solution at the previous load step, admissible perturbations for the second order problem (27) are all the $y - y_- \in K_0^+$. In this way, the set of perturbations is enlarged. It includes all sufficiently smooth functions β which cancel the first order term, without restriction on the sign provided that $\alpha_t(x) + \beta(x) - \alpha_-(x) \geq 0$ for all $x \in (0, 1)$. Consequently, the space of perturbation allows for (small) perturbations β which can be negative at points $x \in (0, 1)$ where $\alpha_t(x) > \alpha_-(x)$. Such a space is a vector space and the eigen-problem can be solved by standard methods of linear algebra by projecting the Hessian to the set of active constraints, cf [1]. Conversely, the stability problem (27), a constrained eigenvalue problem in a convex cone, is of a different nature altogether due to the different topology of the underlying space of variations. The associated discrete problem can be numerically solved by exploiting the orthogonality between the set K_0^+ and its dual $K^* := \{y \in H^1(0, 1) : \langle x, y \rangle \leq 0, \forall x \in K_0^+\}$, cf. [56, 60].

4 Numerical solutions

{sec:numerics}

4.1 Identification of equilibrium branches for the reversible case^{ALB}

We now turn our attention to the solutions beyond the trivial homogeneous branch to identify all the equilibrium branches corresponding to the inhomogeneous **localised**^{ALB} solutions. Our goal is to construct an *equilibrium map* that represents all stable and unstable equilibrium states as a function of the external load [59]. To do so, we use the pseudo-arclength continuation technique implemented in the software AUTO [25], see also [59]. It uses collocation with Lagrange polynomials to discretise the boundary-value problem, in our simulations we had 300 mesh points with 5 collocation nodes and activated mesh adaptation. Then, ^{ALB} it solves the nonlinear equations 9 (and 17 in the compliant case), with the **load and displacement**^{ALB} $\bar{\epsilon}_t$ treated as a continuation parameter.

In order to assess the stability of equilibrium branches we numerically evaluate of the smallest eigenvalue of the second variation by discretizing the integrals (14) and (18) to construct the stiffness matrices $\mathbf{H}$ and $\tilde{\mathbf{H}}$, investigating numerically the sign of the minimal eigenvalue λ_t of the corresponding discrete quadratic form [71]. The finite element discretization of the displacement and damage field (u, α) with n_u displacement degrees-of-freedom $\mathbf{u} = \{u_1, \dots, u_{n_u}\}^T$ and n_α damage degrees-of-freedom $\boldsymbol{\alpha} = \{\alpha_1, \dots, \alpha_{n_\alpha}\}^T$ is given by $u(\mathbf{x}) \approx u_{\text{FE}}(\mathbf{x}) := \sum_{i=1}^{n_u} \mathcal{N}_i^{(u)}(\mathbf{x}) u_i$ and $\alpha(\mathbf{x}) \approx \alpha_{\text{FE}}(\mathbf{x}) := \sum_{i=1}^{n_\alpha} \mathcal{N}_i^{(\alpha)}(\mathbf{x}) \alpha_i$ where $\mathcal{N}^{(u)}(\mathbf{x})$ and $\mathcal{N}^{(\alpha)}(\mathbf{x})$ are the finite element basis functions and u_i^{ALB} and α_i^{ALB} are nodal values for the displacement and damage fields, respectively. **Because both damage and displacement fields are discretised using the same finite element basis functions, we henceforth denote them by the same symbol $\mathcal{N}_i$ for notational convenience.**^{ALB}

Using a hat $\hat{\mathcal{N}}_j(\xi)$ to denote the basis functions defined on the isoparametric domain $\xi \in [-1, 1]$, where j is the local node index within the reference element, the assembly process through which the local basis functions $\hat{\mathcal{N}}_j(\xi)$ are connected to the global finite element fields is the mapping

$$u_{\text{FE}}(x) = \sum_{i=1}^{n_u} \mathcal{N}_i(x) u_i = \sum_{e=1}^{n_{\text{ele}}} \sum_{j=1}^3 \hat{\mathcal{N}}_j(\xi) u_j^e,$$

where u_j^e is the value at the j -th node of the restriction of the field u to the element e , and $u_{\text{FE}}(x)$ is the field in the global physical domain.^{ALB}

For quadratic one-dimensional finite elements with 3 nodes, shape functions $\mathcal{N}_i(x)$ at a node i are given by $\mathcal{N}_1(\xi) = -0.5\xi(1 - \xi)$, $\mathcal{N}_2(\xi) = -0.5\xi(1 + \xi)$ and $\mathcal{N}_3(\xi) = -(1 - \xi)(1 + \xi)$, where ξ is the isoparametric coordinate. The discrete solution $u(x_i)$ provided at discrete nodes x_i by AUTO was first interpolated using B-spline basis functions of degree 3 [31], and then used to calculate the integrals (14) and 18 employing a three-point Gauss integration scheme. The fixed boundary conditions were enforced by removing the rows and columns corresponding to $x = 0$ and $x = 1$ from the stiffness matrices $\mathbf{H}$ and $\tilde{\mathbf{H}}$, **which are real, symmetric, block matrices of dimensions $n_u - 2 + n_\alpha$ and $2(n_u - 2) + n_\alpha$, respectively**^{ALB}. The explicit form of the stiffness matrix for the stiff substrate model is given by

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{uu} & \mathbf{H}_{u\alpha} \\ \mathbf{H}_{\alpha u} & \mathbf{H}_{\alpha\alpha} \end{bmatrix} = \begin{bmatrix} \int_0^1 [\frac{1}{\Lambda^2} \mathcal{N}_i \mathcal{N}_j + (1 - \alpha)^2 \mathcal{N}'_i \mathcal{N}'_j] dx & -2 \int_0^1 (1 - \alpha) u' \mathcal{N}'_i \mathcal{N}_j dx \\ -2 \int_0^1 (1 - \alpha) u' \mathcal{N}_i \mathcal{N}'_j dx & \int_0^1 [(2 + u'^2) \mathcal{N}_i \mathcal{N}_j + \ell^2 \mathcal{N}'_i \mathcal{N}'_j] dx \end{bmatrix}, \quad (28) \quad \{\text{eq:stiffness1}\}$$

whereas for the compliant substrate model it reads

$$\tilde{\mathbf{H}} = \begin{bmatrix} \tilde{\mathbf{H}}_{uu} & \tilde{\mathbf{H}}_{u\alpha} & \tilde{\mathbf{H}}_{uv} \\ \tilde{\mathbf{H}}_{\alpha u} & \tilde{\mathbf{H}}_{\alpha\alpha} & \tilde{\mathbf{H}}_{\alpha v} \\ \tilde{\mathbf{H}}_{vu} & \tilde{\mathbf{H}}_{v\alpha} & \tilde{\mathbf{H}}_{vv} \end{bmatrix} = \begin{bmatrix} \int_0^1 [\frac{1}{\Lambda^2} \mathcal{N}_i \mathcal{N}_j + (1 - \alpha)^2 \mathcal{N}'_i \mathcal{N}'_j] dx & -2 \int_0^1 (1 - \alpha) u' \mathcal{N}'_i \mathcal{N}_{j'}^{\text{ALB}} dx & - \int_0^1 \rho_l \mathcal{N}'_i \mathcal{N}_{j'}^{\text{ALB}} dx \\ -2 \int_0^1 (1 - \alpha) u' \mathcal{N}_{i'}^{\text{ALB}} \mathcal{N}'_j dx & \int_0^1 [(2 + u'^2) \mathcal{N}_{i'}^{\text{ALB}} \mathcal{N}_{j'}^{\text{ALB}} + \ell^2 \mathcal{N}'_{i'} \mathcal{N}'_{j'}^{\text{ALB}}] dx & \\ - \int_0^1 [\frac{1}{\Lambda^2} \mathcal{N}_{i'}^{\text{ALB}} \mathcal{N}_j] dx & 0 & \int_0^1 \rho_l \mathcal{N}_{i'}^{\text{ALB}} \mathcal{N}_j dx \end{bmatrix} \quad (29) \quad \{\text{eq:stiffness2}\}$$

where $i, j = 0, \dots, n_u; i', j' = 0, \dots, n_\alpha$, and $k, l = 0, \dots, n_v$ are the degrees of freedom indices for the **film displacement, damage, and foundation displacement, respectively.**^{ALB} Our objective is to establish a branch switching strategy that, as the external loading parameter monotonically increases, allows equilibrium branch transitions when the current branch ceases to be stable or to exist. This strategy must ensure the system's re-stabilization following an instability in a dissipative manner. In a quasi-static scenario, it should select a new locally stable equilibrium branch with inherently lower energy. Considering applications in structural mechanics, our approach to selecting the new equilibrium branch relies on a criterion of local energy minimization (LEM) which emulates the zero viscosity limit of overdamped viscous dynamics. This approach differs from a global energy minimizing (GEM) strategy,

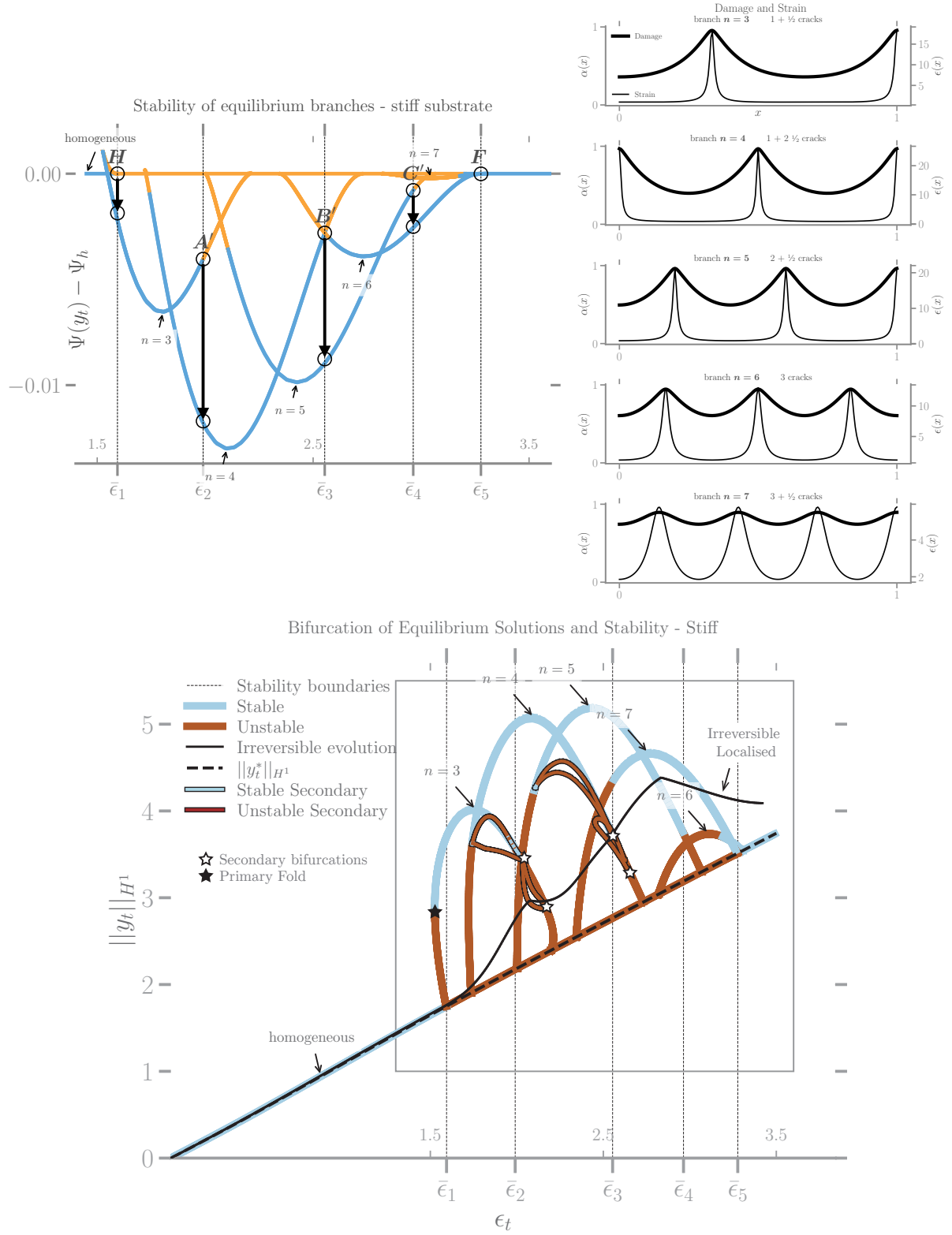


Figure 2: Equilibrium branches for phase-field thin film model on rigid elastic foundation. Left: energy difference $\Delta\Psi$ between current and homogeneous configurations at equal load. Stability color-coded: blue (stable), orange (unstable) based on smallest eigenvalue of stiffness matrix H . Branches parametrised by integer n ; arrows indicate branch switching at stability loss. Right: damage and strain profiles of minimum energy configurations on each branch. Bottom: bifurcation and stability map.^{OUS}

{fig:branches

which may be more relevant in biomechanical applications [67]. According to LEM protocol, during quasi-static loading, the system will remain in a metastable state (a local minimum of energy) until it becomes unstable. Subsequently, during an isolated switching event, the new equilibrium branch will be chosen using a descent algorithm [62].

Fig. 2, 3 show equilibrium branches that solve the nonlinear equations 9 and 17. These branches are represented in the top-left subfigure^{ALB} by plotting the energy difference $\Delta\Psi$ between the energy of the current solution and the energy of the homogeneous solution $\Psi^{\text{hom}}(\bar{\epsilon}_t)$ at the current value of the loading parameter $\bar{\epsilon}_t$. The stability of solutions is color-coded, light blue and orange representing stable and unstable solutions, respectively. Stability is discerned through numerical evaluation of the smallest eigenvalue λ_t of the stiffness matrix $\mathbf{H}$ (and $\tilde{\mathbf{H}}$) defined in (28) (respectively, in (29)). Because eigenvalues represent the local curvatures of the energy functional, the sign of the smallest λ_t determines the stability of the solution, with $\inf \lambda_t > 0$ ^{ALB} indicating stability and $\inf \lambda_t < 0$ indicating instability.

Under the LEM protocol, the system explores the fundamental branch where the homogeneous solution is stable until $\bar{\epsilon}_1$, identified using linear stability analysis, see Figure 2. At the critical load on the homogeneous branch at point H , a first instability determines a branch switching transition, from the trivial branch with wavenumber $n = 0$ to the nontrivial equilibrium branch with $n = 3$. The ensuing non-homogeneous configuration is linearly stable, as seen in Fig. 2. We depict the lowest energy configuration on this equilibrium branch in Fig. 2 at point A , consisting of two simultaneously nucleated localised cracks, one inside the domain and one on the boundary.

When the loading parameter $\bar{\epsilon}_t$ is further increased, the equilibrium configuration with $n = 3$ loses linear stability at point A' . In Fig. 2 we represent state transitions according to the LEM protocol by arrows. It can be observed that under this protocol, the only available transition from point A' is to the branch with $n = 4$, which is locally stable within at the corresponding applied strain $\bar{\epsilon}_2$. The minimum energy configuration on the branch with $n = 4$ is depicted in Fig. 2-right at point B , consisting of two cracks: one inside the domain and two on the right and left boundaries.

Further increasing the load, the stability of the branch with $n = 4$ is lost at point B' , a single available transition leads the system to the branch with $n = 5$. While the branch with $n = 6$ appears accessible due to its lower energy compared to the current state, it is unstable at the current value of $\bar{\epsilon}_3$. The lowest energy configurations on the branches with $n = 5$ and $n = 6$ are illustrated in Fig. 2(right). Increasing the load along the branch with $n = 5$ which loses linear stability at point C' , a branch switching event occurs towards the branch with $n = 7$, the corresponding minimum energy configuration on this branch is depicted in Fig. 2(right). Finally, this branch reconnects to the homogeneous branch with $n = 0$ at point F . Remark that, in the case of a stiff substrate, the LEM protocol identifies a unique evolution path, with the system having a single bifurcation option, or available branch, at each instability point.

For the compliant substrate model, the equilibrium branches that solve the nonlinear equations (Eq. 17) are shown in Fig. 3. We again plot the energy difference $\Delta\tilde{\Psi}$ between the energy of the current solution and the energy of the homogeneous solution $\Psi^{\text{hom}}(\bar{\epsilon})$ at the current value of the loading parameter $\bar{\epsilon}$. The stability of solutions is color coded (light blue indicates *stable* states, orange identifies *unstable* states), the stability marker being given by the numerical evaluation of the smallest eigenvalue of the stiffness matrix $\tilde{\mathbf{H}}$, defined in (29).

According to the LEM protocol with a loading history starting in the unloaded and sound configuration, the initial transition from the trivial solution to the only available branch with $n = 2$ occurs at $\bar{\epsilon}_1$, marked as point H . As illustrated in Fig. 3 (right), the damage and strain profiles of the lowest energy configuration at point A is characterised by two boundary cracks. As the loading increases, the system persists on the $n = 2$ branch until point A' which marks an instability. The system can now access two equilibrium branches: $n = 3$ and $n = 4$ shown in gray and black arrows. Fig. 3 (right) shows the typical damage and strain profiles on these branches. Branches $n = 3$ and $n = 4$ feature one bulk crack plus one boundary crack, and one bulk crack plus two boundary cracks, respectively.

The choice of the subsequent branch transition at point A' will dictate the ensuing crack growth. On branches with $n = 3$ and $n = 4$, at the instability points B' and C' , the system will once again encounter two available equilibrium branches. From $n = 3$ branch at point B' , the system can transition to either the $n = 4$ or $n = 5$ branch. Opting for the $n = 4$ branch, the subsequent branch selection occurs at point C' , offering branches with $n = 6$ or $n = 5$. Notably, the $n = 5$ branch smoothly reconnects with the trivial solution at point F . However, the $n = 6$ branch experiences another instability at point D' before rejoining the trivial solution.

Diagrams 2.bottom and 3.bottom (for the compliant system) offers further qualitative insight of the bifurcation behaviour of the system. With reference to the stiff system (considerations in the compliant case are, at first order, analogous), the system evolves starting from the zero state along the stable

[ALB 3] check notation, maybe $\inf_i \lambda_t^{(i)}, \lambda_{ti}, \lambda_{it}$

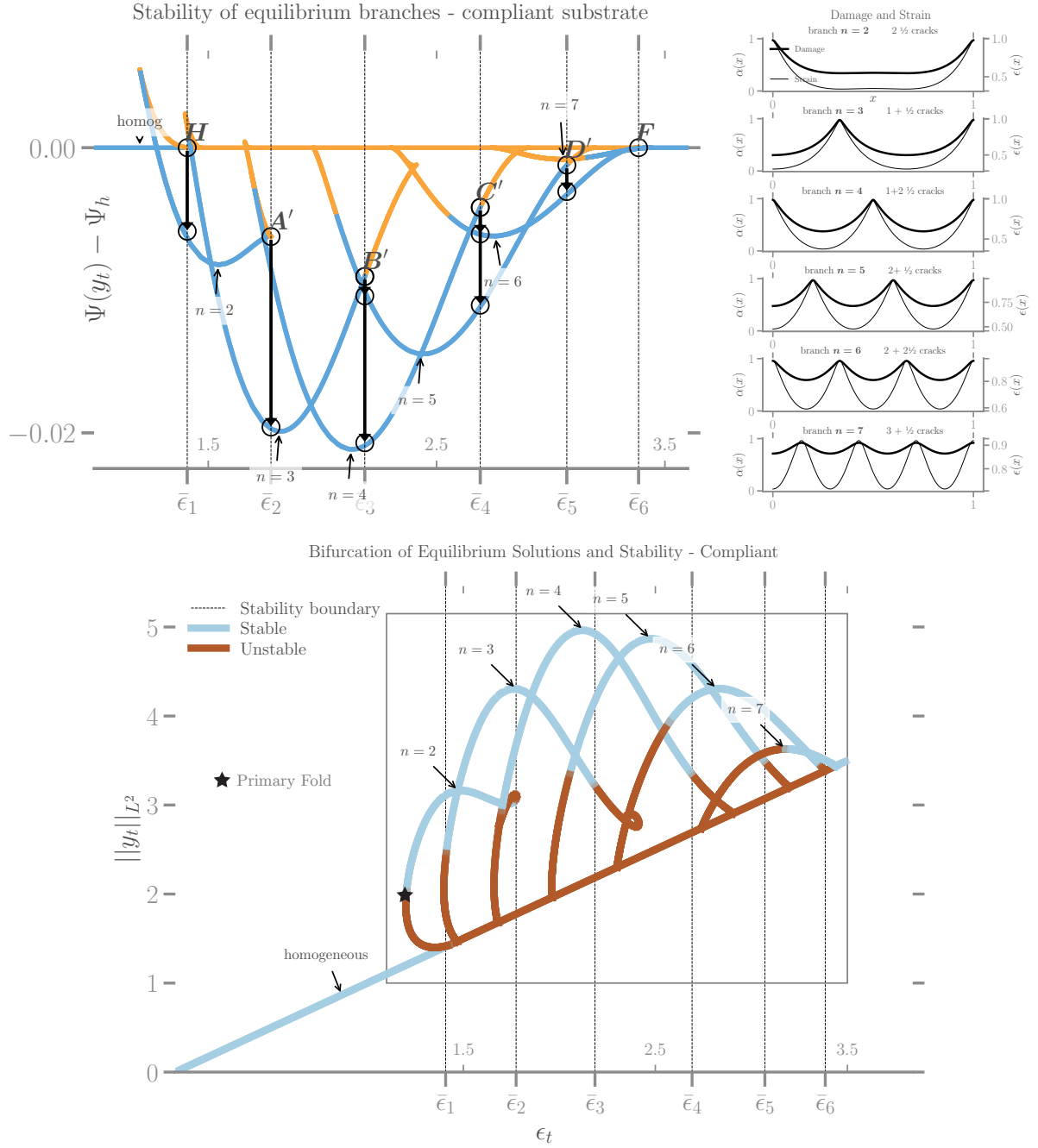


Figure 3: Equilibrium branches for phase-field thin film model on compliant elastic foundation. Left: energy difference $\Delta\Psi$ between current and homogeneous configurations at equal load. Stability color-coded: blue (stable), orange (unstable) based on smallest eigenvalue of stiffness matrix H . Branches parametrised by integer n ; arrows indicate branch switching at stability loss. Right: damage and strain profiles of minimum energy configurations on each branch. Bottom: bifurcation and stability map.^{OUS}

{fig:branches

homogeneous branch, exhibiting a subcritical pitchfork bifurcation at load ϵ_1 (indicated as point H). Pitchfork bifurcations usually occur in problems with symmetry and symmetry breaking, which is the case for the homogeneous solution curve. In this case, this corresponds to the loss of stability for the reversible system. The bifurcation is subcritical as the fundamental branch, unstable beyond ϵ_1 , shows the existence of a bifurcating branch with smaller values of the control parameter, ϵ_t . Whilst several other bifurcations from the homogeneous branch occur (at the $\epsilon_i, i \leq 5$) without change in stability, the homogeneous solution eventually becomes stable again at the (supercritical) pitchfork point F , at load ϵ_5 . Following the first emergent branch at ϵ_1 , the system explores higher energy states than the corresponding homogeneous states, thus the unstable character (cf. energy diagram in Figure 2-top left). The solution curve folds back at the turning point X , as stability is restored. The ensuing stability switch at the fold is characterised by a vertical tangent to the solution curve where its slope reverses and the two solution branches (unstable and stable) connect at the turning point. Consequently, the Hessian is there singular with one eigenvalue passing from negative to positive with an associated change of stability yet without implying further (secondary) solution branches. Further along the bifurcated and localised branch $n = 2$, two other bifurcations occur leading to the emergence of two new secondary branches which stem from secondary bifurcation points, highlighted with a black border against the primary bifurcated branch. Note that, whilst the first secondary branch shows a stable part and a (corner) fold without change of stability, the second secondary branch is entirely unstable. Similarly to what occurs for the homogeneous branch, bifurcations may occur without change of stability, signaling the onset of zero eigenvalues in the presence of other negative ones. The bifurcation diagram of the compliant system (Figure 3.bottom) shows a similar behaviour, with additional complexity due to the presence of additional degrees of freedom associated to the substrate deformation, leading to a richer bifurcation structure with additional turning points with and without change of stability. ^{ALB}

For the current choice of material parameters, the compliant substrate model shows more branch switching events and notably, ^{ALB} non-uniqueness of the global response. Indeed, differently from ^{ALB} the stiff substrate model where ^{ALB} the LEM strategy identifies a unique evolution path, while ^{ALB} in the compliant substrate model the system can follow *eight* different trajectories in phase-space, given by the various available path-bifurcation choices. This additional richness provides a good test case for numerical optimization methods, which we will discuss in the following.

4.2 Equilibrium branch selection by overdamped viscous dynamics

We consider now the equilibrium solutions reachable through the line-search based quasi-Newton algorithms such as conjugate gradient or the BFGS optimization, which effectively mimic the zero viscosity limit of overdamped viscous dynamics [70]. Quasi-Newton methods serve as alternatives to Newton's method for locating roots or local extrema of functions. Particularly advantageous in scenarios where computing the Hessian at each iteration is impractical or computationally expensive, these methods circumvent the need for explicit computation of energy derivatives. Instead, they rely on evaluating the function value and its gradient and updating the Hessian by analyzing successive gradient vectors.

Quasi-Newton methods are highly suitable for solving the phase-field equation of fracture, particularly when compared to standard Newton method-based monolithic solvers. Such solvers, which simultaneously solve the equations for both damage and displacement variables, often falter when confronted with nonconvex energy functionals. For example, as demonstrated in [84], the Newton method-based monolithic algorithm does not consistently handle brittle fracture scenarios involving abrupt crack propagation. Recently, quasi-Newton methods, particularly the BFGS variant, have been employed to effectively solve the system of coupled governing equations in a monolithic fashion within the phase-field method of fracture. These methods have demonstrated success in various engineering applications, as evidenced by [41, 86, 67, 48].

Our primary objective is not to provide a comprehensive assessment of quasi-Newton methods on a global scale. Instead, our focus lies in scrutinizing their behavior and performance specifically concerning equilibrium branch selections within our simplified framework, where all branches are readily identified. By narrowing our scope to this specific aspect, we aim to gain insights into the effectiveness and reliability of quasi-Newton methods in reaching stable states of our system.

Recall that quasi-Newton algorithms only evaluate the function value and its gradient to reach equilibrium configurations. This implies that in our case, we need to evaluate integrals (3) and (4), along with their first variations given by (7) and (16), in the stiff and compliant models, respectively. We discretised the integrals (7) and (16) to construct the residuals vectors using finite elements to obtain

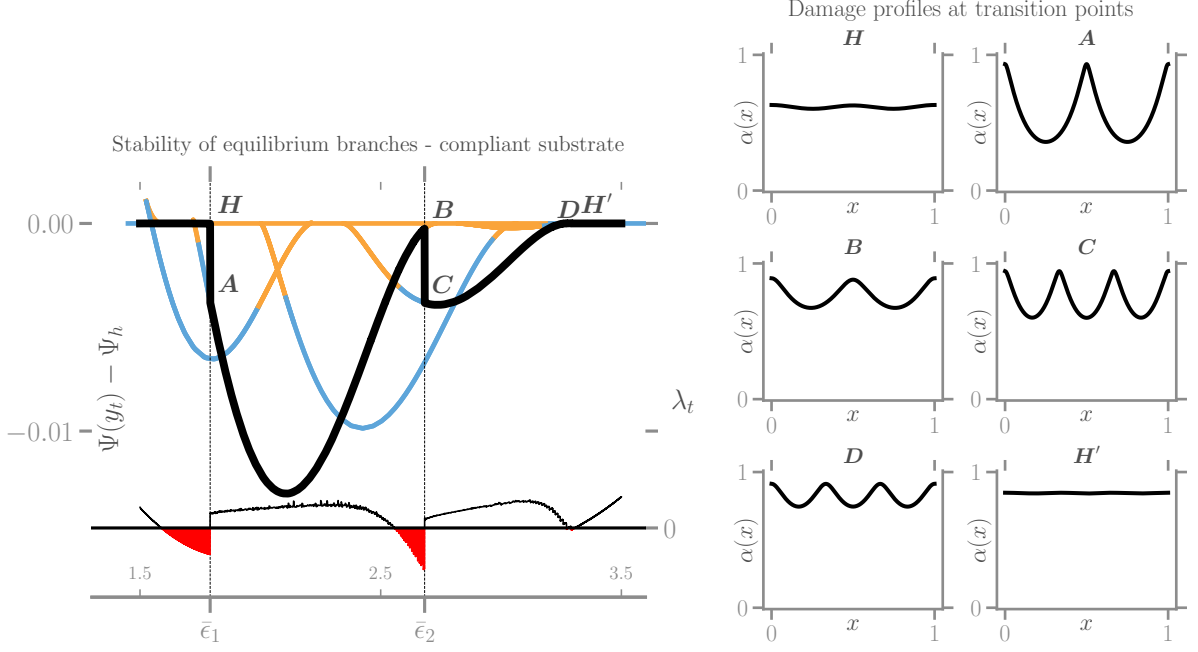


Figure 4: Stiff substrate. Quasi-static L-BFGS simulations: (a) energy difference $\Delta\Psi$ between quasi-Newton and homogeneous solutions superimposed on equilibrium branches (light blue: stable; orange: unstable). Bottom: smallest eigenvalue λ_t of second variation Ψ'' vs. loading parameter $\bar{\epsilon}$. Red indicates unstable region ($\lambda_t < 0$) where solver solutions violate evolution law. Right: damage profiles at transition endstates.^{OUS}

{fig:tempo1}

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_u \\ \mathbf{R}_\alpha \end{bmatrix} = \begin{bmatrix} \int_0^1 [(1-\alpha)^2 u' \mathcal{N}'_i + \frac{1}{\Lambda^2} (u - \tilde{u}) \mathcal{N}_i] dx \\ \int_0^1 [(\frac{1}{2} u'^2 a'(\alpha) + w'(\alpha)) \mathcal{N}_j + \ell^2 \alpha' \mathcal{N}'_j] dx \end{bmatrix}. \quad (30) \quad \{\text{residual1}\}$$

^{ALB} where $i = 1, \dots, n_u - 1$, and $j = 0, \dots, n_\alpha$ in the stiff model and

$$\tilde{\mathbf{R}} = \begin{bmatrix} \mathbf{R}_u \\ \mathbf{R}_\alpha \\ \mathbf{R}_{\tilde{u}} \end{bmatrix} = \begin{bmatrix} \int_0^1 [(1-\alpha)^2 u' \mathcal{N}'_i + \frac{1}{\Lambda^2} (u - \tilde{u}) \mathcal{N}_i] dx \\ \int_0^1 [(\frac{1}{2} u'^2 a'(\alpha) + w'(\alpha)) \mathcal{N}_j + \ell^2 \alpha' \mathcal{N}'_j] dx \\ \int_0^1 [\rho \tilde{u}' \mathcal{N}'_i + \frac{1}{\Lambda^2} (u - \tilde{u}) \mathcal{N}_i] dx \end{bmatrix}. \quad (31) \quad \{\text{residual2}\}$$

^{ALB} where $i, k = 1, \dots, n_u - 1$, and $j = 0, \dots, n_\alpha$ in the compliant substrate model.

Among iterative methods for large-scale unconstrained optimization, particularly when dealing with possibly dense Hessian matrices, quasi-Newton methods often emerge as preferable alternatives to the widely-used Newton-Raphson (NR) algorithm. The NR algorithm, conventionally utilised for solving linear equations to determine the correction $\Delta \mathbf{X}^{(k)}$ from the current estimate $\mathbf{X}^{(k)} = (\mathbf{u}^{(k)}, \boldsymbol{\alpha}^{(k)})$ at iteration k , is expressed in our context as:

$$K_{ij} \Delta X_j^{(k)} + R_i = 0, \quad (32) \quad \{\text{Eq:NR}\}$$

where the discrete stiffness matrix $\mathbf{H}$ and bulk forces $\mathbf{R}$ are computed with the initial guess $\mathbf{X}^{(k)}$. Subsequently, the guess is updated as $\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} + \Delta \mathbf{X}^{(k)}$ after solving Equation (32) using LU factorization [71]. It's evident that the NR algorithm fails if the discrete stiffness matrix $\mathbf{H}$ isn't invertible.

On the other hand, quasi-Newton methods are well-established (see standard textbooks, e.g., [1, 58]), and generate a sequence $\{\mathbf{X}^{(k)}\}$ according to the following scheme:

$$\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} + h^{(k)} \mathbf{p}^{(k)}, \quad k = 0, 1, \dots \quad (33)$$

with

$$\mathbf{p}^{(k)} = -(\mathbf{B}^{(k)})^{-1} \mathbf{R}, \quad (34)$$

where $(\mathbf{B}^{(k)})^{-1}$ approximates the inverse of the Hessian matrix $\mathbf{H}$ and $h^{(k)}$ represents a step length. Particularly, instead of computing $(\mathbf{B}^{(k)})^{-1}$ at each iteration k , these methods update $(\mathbf{B}^{(k)})^{-1}$ in a

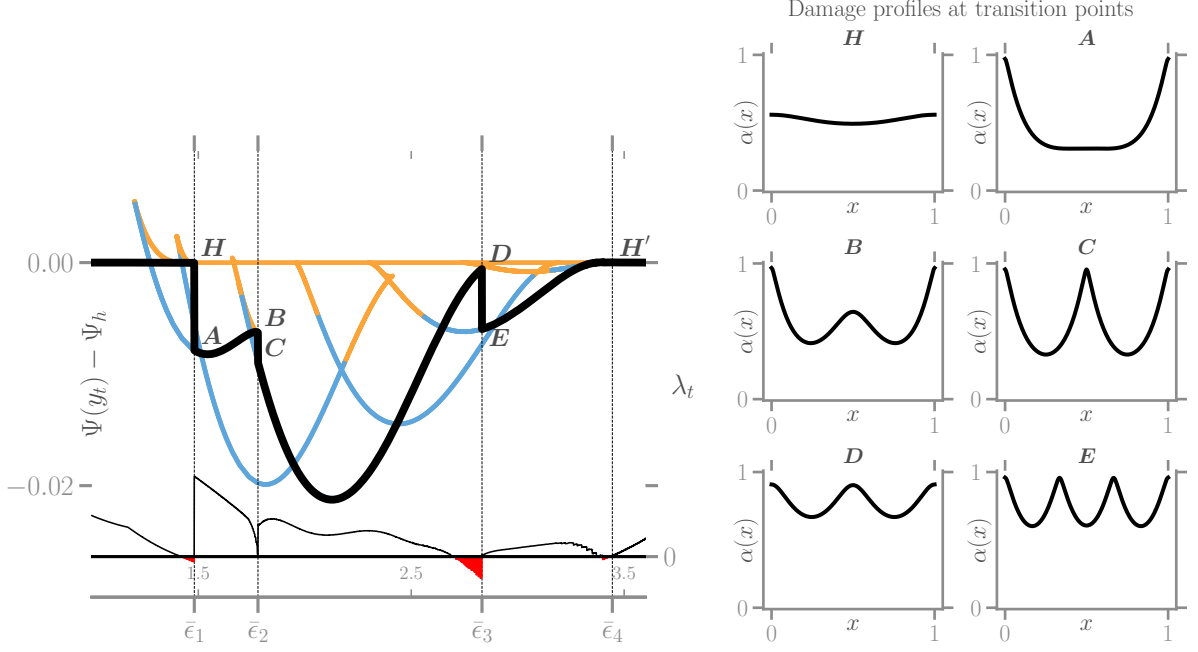


Figure 5: Compliant substrate. Quasi-static L-BFGS simulations: (a) energy difference $\Delta\Psi$ between quasi-Newton and homogeneous solutions superimposed on equilibrium branches (light blue: stable; orange: unstable). Bottom: smallest eigenvalue λ_t of second variation Ψ'' vs. loading parameter $\bar{\epsilon}$. Red indicates unstable region ($\lambda_t < 0$) for equilibrium solution. Right: damage profiles at transition endstates.^{OUS}

{fig:tempo2}

straightforward manner to obtain the new approximation $(\mathbf{B}^{(k+1)})^{-1}$ for the subsequent iteration. Additionally, rather than storing full dense $N \times N$ approximations, they only retain a few vectors of length $N = n_\alpha + n_u$, enabling implicit representation of the approximations. Moreover, the choice of the step length $h^{(k)}$ is carried out through a line search to minimise a function $f(h) = f(\mathbf{X}^{(k)} + h\mathbf{p}^{(k)})$ in order to find an acceptable step size $h^{(k)}$ such that $h^{(k)} \in \arg \min_h f$.

Among quasi-Newton schemes, the L-BFGS method is widely regarded as one of the most efficient and well-suited for large-scale problems due to its limited and user-controlled storage requirements. This method relies on constructing an approximation of the inverse Hessian matrix, leveraging curvature information solely from recent iterations. Note also that the update formula for the approximative in successive minimization steps depends on the adapted algorithm. These algorithms are extensively used in the literature, and we refer the reader to the references for a more detailed description [51, 87, 1, 58, 73, 45, 22]. However, quasi-Newton methods do present certain drawbacks, notably slow convergence for ill-conditioned problems, particularly when the eigenvalues of the Hessian matrix are widely dispersed [73].

In our numerical experiments, we employ the BFGS and L-BFGS solvers from the Python SciPy library [81] and the Alglib library [12]. These solvers utilise the residual vectors (see 30 and 31) at each finite element node, alongside the values of integrals (3) and (4). The following results were obtained using the Alglib library [12]; however, our findings remain consistent across the other libraries mentioned.

In Figure 4, we present the results of our numerical experiments, overlaying solutions obtained with the quasi-Newton algorithm onto the equilibrium map determined using the pseudo-arclength continuation method described in the previous section. Fig. 4-bottom shows the smallest eigenvalue λ_t whose positivity informs on the stability of the solution. All the branch switching events are delayed in the sense they do not take place when the smallest eigenvalue of the second variation vanishes but only beyond the loads corresponding to the eigenvalue's sign transition. Notably, the expected first branch switching event—from the homogeneous solution to the branch with $n = 3$ as outlined in the LEM protocol—did not occur at point H as anticipated. During a monotonic loading, the system remains on the trivial branch, as shown in Fig. 2(a), until it transitions to the branch with $n = 4$ at a value of the load much higher than the critical loading parameter $\bar{\epsilon}_1$. By monitoring the smallest eigenvalue of the second energy variation at the solution, we observe that indeed the solutions computed via the quasi-Newton

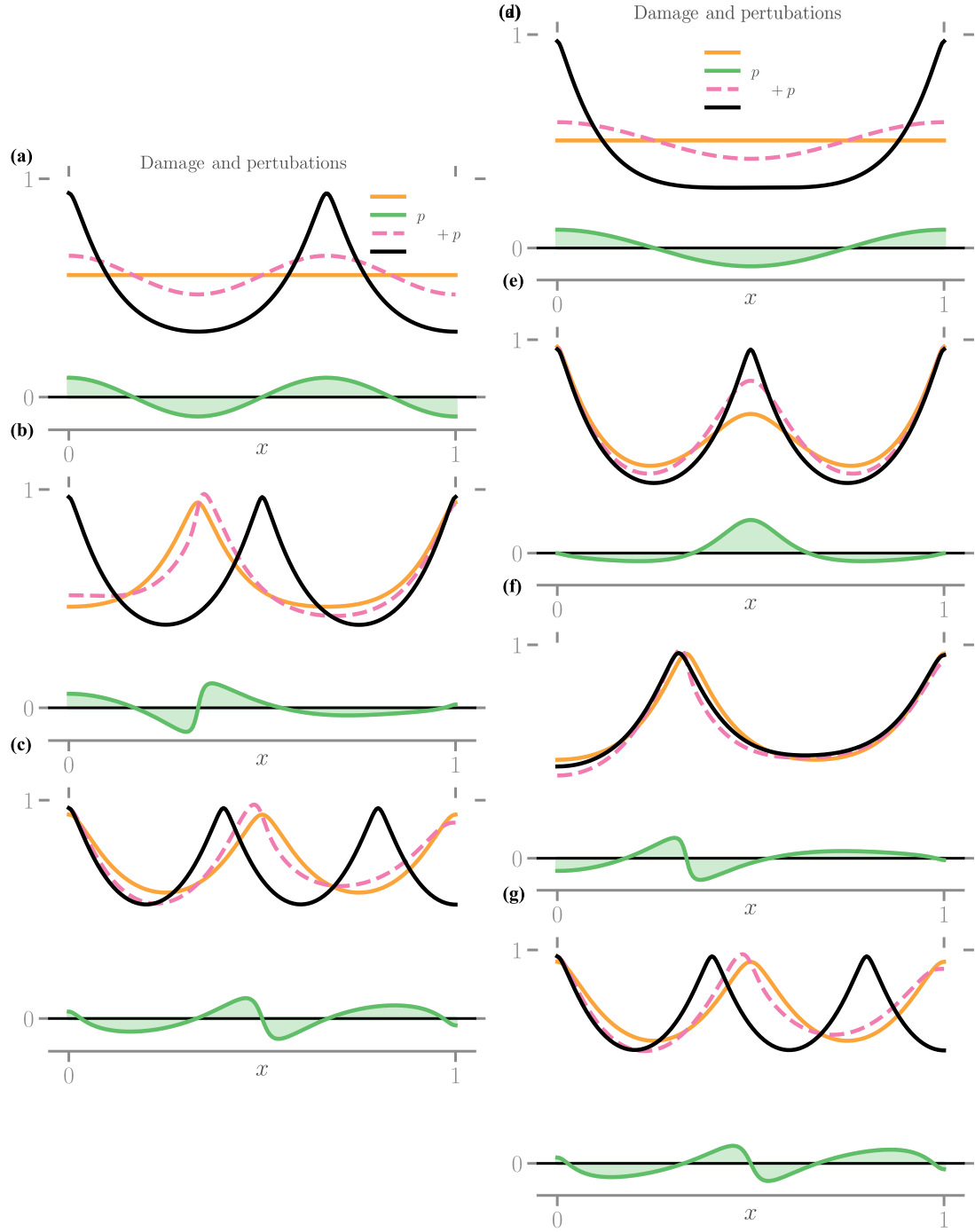


Figure 6: Profiles of damage fields and perturbations showing kick algorithm effect in state transitions (unstable to stable). Branch switching events correspond to load values $\bar{\epsilon}_i$ with $i \in \mathbb{N}$ in Figs. 7-8. Left column: stiff substrate model; right column: compliant substrate model. Orange line: unstable damage field α^* ; green line: eigenvector p associated with smallest eigenvalue (positive/negative values highlighted). Dashed line: perturbed damage field $\alpha^* + p$ used as initial guess; solid black line: solution α_t after first-order convergence.^{OUS}

{fig:kick}

solver are unstable beyond $\bar{\epsilon}_1$, which is consistent with the linear stability results for the trivial branch. A closer inspection of the solution field reveals that the quasi-Newton solutions are not homogeneous but instead exhibit a small perturbation akin to the instability mode calculated analytically (i.e., the eigenvector associated with the smallest eigenvalue). For instance, the quasi-Newton solution before the first branch switching event is shown in inset (state H) in Fig. 4-right, where we see that the equilibrium damage shows a slight oscillation reminiscent of the eigenmode $\alpha_n(x) \sim \cos(n\pi x)$, with $n = 3$. The state transition captured by the quasi-Newton method is the one towards the $n = 4$ branch, with one bulk crack and two boundary cracks, see inset (A) in Fig. 4-right. If the transition had taken place at $\bar{\epsilon}_1$ as anticipated by our LEM strategy, the system would have reached a configuration with two bulk cracks and one boundary crack, as shown in Fig. 2-right (branch $n = 5$).

With continued loading, the damage increases and the system evolves along the $n = 4$ branch beyond the loading value at which stability is lost, as shown in Fig. 4-right. A dissipative branch switching event takes place from the current branch ($n = 4$) at point B' to the branch with $n = 6$; see Fig. 4(right) and the inset C for the corresponding damage profile. The delay in bifurcation results in a branch selection event different from the one in our LEM protocol. The delays at bifurcation for the first and second branch switching events leads to a completely different evolutionary path compared to the LEM protocol until it finally reconnects with the homogeneous branch at point C' .

For the compliant substrate model, the solutions of our numerical experiments obtained from the quasi-Newton algorithm (black thick line) are overlaid in Fig. 5(left) onto the equilibrium map, while their stability is depicted in Fig. 5-bottom. Notably, we observe a delay, albeit less pronounced compared to the stiff substrate model, for the bifurcation from the trivial branch which occurs at a value higher than the analytically determined critical load $\bar{\epsilon}_1$. Interestingly, the quasi-Newton method switches to the branch with $n = 2$, which exhibits lower energy compared to the branch with $n = 3$. This occurs despite the delay in bifurcation because the branch $n = 3$ remains accessible for the current value $\bar{\epsilon}$. It's worth noting that the LEM protocol also anticipated a first branch selection event leading to the branch with $n = 2$ with two boundary cracks. We again observe that, even before the first branch-changing event, the quasi-Newton solution deviates from the homogeneous branch, and the damage profile has is perturbed by an oscillatory term of the form $\cos(n\pi x)$, reminiscent of the eigenmode α_n with $n = 2$. These perturbations of the equilibrium profiles are negligible from the global energetic standpoint, as it can be inferred in Figure 5(a), by the superposition between the energy of the computed evolution and the exact total energy of the homogeneous solution, yet have a role in the selection of the next branch as will be evidenced in the following.

The system moves along the branch with $n = 2$ as the load increases, until the stability transition at load $\bar{\epsilon}_2$, indicated by point A' . Energy minimization brings the system to the branch $n = 4$ although the branch with $n = 3$ is also accessible and has lower energy. Figure 5(right), displays the corresponding damage profiles before and after the branch switching events at points B' and C . The final transition before the reconnection to the homogeneous branch takes place at point C' , which is also an unstable configuration. The transition brings the system to the branch $n = 6$ as seen in Fig. 5(right), the corresponding stable damage profile at point D is shown in Fig. 5(right).

In summary, while the results of the quasi-Newton minimization simulations exhibit overall similarities in both models, the trajectory taken by the system in the compliant model more closely adheres with the LEM protocol outlined above. This alignment can be attributed to the specific energy landscape of the compliant model. In case of indeterminacy of the trajectory, such as when multiple stable solutions exist at a loss of stability, how to identify an evolution direction?

4.3 Hybrid *kick* algorithm for branch switching

Our investigation shows that the solutions generated by the quasi-Newton algorithms deviate from the LEM protocol outlined in the preceding sections. We observe delayed bifurcations and the persistence of unstable solutions, both affecting branch selection events along the evolution. This is indeed related to the fact that the energy landscape is already flat when the determinant Hessian gets close to zero. Testing the energy expansion (2) along a the direction of a bifurcation mode, that is setting $y - y_t = hp_n$, where $p_n := (v_n, \alpha_n)(x)$ is the n -th eigenmode associated with the Hessian and $0 < h \in \mathbb{R}$, we obtain that, for admissible states y δ -close to the equilibrium y_t we can write the following lower bound

$$\Psi(y) - \Psi(y_t) \geq \frac{h^2}{2} \lambda_t \|p_0\|^2, \quad \forall y : \|y - y_t\| \leq \delta,$$

where λ_t indicates the smallest eigenvalue and w_0 the associated eigenmode. The first order term $\delta\Psi(y_t)(v_n, \alpha_n)$ vanishes identically owing to the fact that bifurcation modes are admissible fields for the

equilibrium condition. When the smallest eigenvalue continuously approaches zero, the energy landscape morphs from being locally flat (at first order) and locally convex in all admissible directions including, in particular, the directions associated with the eigenmodes, to losing local convexity in the one non-trivial direction associated with the eigenvalue that has changed sign. In the numerical practice, the convergence of quasi-Newton algorithms depends on the construction of an (approximate) strictly positive Hessian matrix. As a consequence, such approximation systematically rules out the ability to capture the change of sign (of the smallest eigenvalue) of the Hessian, that is its singularity, thus the onset of instability and the instability mode. This can justify the systematic (and algorithm-dependent) delay of the bifurcation events via the quasi-Newton solver, observed in Fig. 4 and 5.

To address this challenge, we introduce a hybrid approach that explicitly takes into account the singular mode of the Hessian. In this method, we continuously monitor the smallest eigenvalue of the complete Hessian matrix for the equilibrium solutions obtained from the quasi-Newton algorithm.

When this eigenvalue significantly diminishes, indicating potential instability, first, we perform a full Newton-Raphson refinement using the solution returned from the quasi-Newton algorithm as an initial guess and obtain the fully homogeneous solution. Then we calculate the corresponding eigenvector $\mathbf{p}$, normalised such that $\|\mathbf{p}\| = 1$. We then use this eigenvector to perturb the current solution $\mathbf{X}^*$. This perturbation sets the initial guess for the next minimization step in the quasi-Newton algorithm as $\tilde{\mathbf{X}}^{(0)} = \mathbf{X}^* + \eta\mathbf{p}$, where η is the step size. This step size can be determined through a line-search algorithm by minimizing the one-dimensional energy slices along the energy descent mode given by the function

$$f(\eta) = \Psi(\mathbf{X} + \eta\mathbf{p}), \quad (35) \quad \{\text{eqn:energy-s}$$

to find an optimal value η , such that $\eta \in \arg \min_{\eta} f$. We then run the quasi-Newton step to obtain a new critical point $\mathbf{X}'$.

The outcome of this hybrid algorithm is the three anticipated branch-switching events identified for the first model previously reported in Fig. 2. For each of these three cases, the snapshots of the damage fields, the associated instability mode $\mathbf{p}$, the perturbed states $\tilde{\mathbf{X}}^{(0)}$, and the converged solutions $\mathbf{X}'$ are shown in Fig. 6 in the right column.

When the quasi-Newton algorithm is initiated using the perturbed state $\tilde{\mathbf{X}}^{(0)}$ as the initial guess, a state transition occurs which leads the system into a qualitatively different state, cf. the final converged solutions returned from the quasi-Newton minimization are shown in Figs. 6. For the stiff substrate model, the instability mode which corresponds to the bifurcation from the trivial branch has the form $\cos(n\pi x)$ with $n = 3$, as identified through the linear stability analysis (see Fig. 6(a)). The perturbed quasi-Newton minimization returns a state with two boundary cracks and one interior crack, as shown in Fig. 6(a), which corresponds to the solution found using the arc-length continuation algorithm on the branch with $n = 3$ (see Fig. 2(b), inset A). The hybrid algorithm is also successful in capturing the next two anticipated branch switching events: from the branch with $n = 3$ to the branch with $n = 4$ (see Fig. 6(b)) and from branch with $n = 4$ to the branch with $n = 5$ (see Fig. 6(c)). The hybrid algorithm is able to capture the unique path dictated by the LEM protocol shown in Fig. 2. We also show the evolution of the smallest eigenvalue during all loading process in Fig. 7-left where it is clear that the solution (and the evolution as a whole) is stable.

Let's now focus on the compliant substrate model. In this case, according to the LEM protocol, the path through which the system may evolve under quasi-static loading is not unique due to the multiplicity of (stable) solutions at critical loads. Note that this is not the case for the first model where a single equilibrium branch is accessible to the system for all given critical loads. Despite the quasi-Newton algorithm was able to reproduce one of the possible paths predicted by the LEM protocol for the compliant substrate model, the branch switching events are delayed with respect to the anticipated bifurcations.

The system's response under the hybrid algorithm is depicted in Fig. 6(d-g). The first column displays the perturbed damage profiles, while the second column shows the outcomes of the energy minimization process starting from this perturbed state. A stable state with two nucleated boundary cracks (branch $n = 2$) is achieved following the destabilization of the trivial branch, as illustrated in Fig. 6(d). Subsequently, an interior crack nucleates in the middle of the system (branch $n = 4$), as shown in Fig. 6(e). The third event involves a transition from branch $n = 3$ to branch $n = 5$. The event depicted in Fig. 6(f) is the transition to branch $n = 6$. The last event before reconnecting to the trivial branch is the transition from branch $n = 6$ to branch $n = 7$, the only branch available to the system at this load, see in Fig. 6(f). To summarise, the branches selected by the hybrid algorithm follow the sequence $0 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 0$, which differs from the sequence $0 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 0$ followed by the quasi-Newton algorithm.

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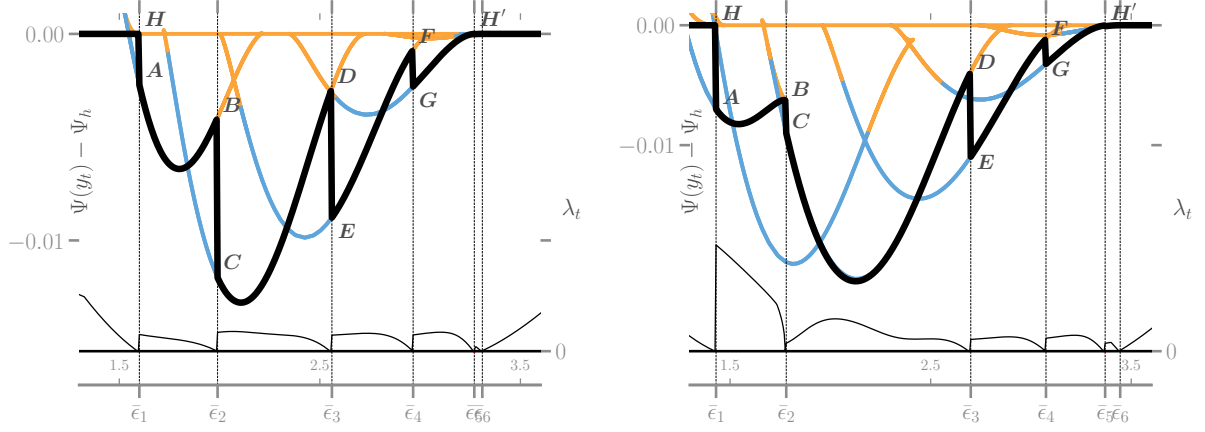


Figure 7: Quasi-static L-BFGS simulations: energy difference $\Delta\Psi$ between quasi-Newton and homogeneous solutions superimposed on equilibrium branches (left and right panels). Bottom: smallest eigenvalue of second variation vs. loading parameter $\bar{\epsilon}$. Positive eigenvalues indicate solution and overall evolution stability.^{OUS}

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Our numerical findings suggest that equilibria close to the Hessian degeneracy (i.e. with small but positive eigenvalues) are robust in the sense that they are insensitive to perturbations. This is highlighted by the fact that running the quasi-Newton solver on a perturbed state close to the Hessian sign transition yields, after convergence, the same equilibrium state prior to perturbation. On the other hand, when the smallest eigenvalue of the system's Hessian is small and negative, the step size η obtained by minimization of (35) is sufficient for the quasi-Newton algorithm to effectively escape the flat region of the energy landscape.

4.4 Irreversible evolutions and stability

The algorithm proposed in the previous section consists in a branch switching strategy, similar to that which has been proposed by [7] by exploiting the concept of the tangent stiffness matrix, obtained by projection of the Hessian on the subset of (linearised) active constraints. The main difference with respect to the approach proposed here is that we now account for the fully nonlinear stability problem at second order, in presence of (weakly active) irreversibility constraints, explicitly representing the nonlinearity of the admissible perturbation space.^{ALB} To highlight the role of constraints on the determination of the evolution and its consequences on the space of perturbations, we discuss the full stability problem inequality (27) in our simplified one-dimensional setting through an illustrative numerical computation. Irreversibility manifests both as a pointwise constraint and as a global nonlinearity in the perturbation space. Firstly, it rules out transitions between branches that require a local decrease of damage in favour of a global decrease of energy (e.g. from branch $n = 3$ to branch $n = 4$, see profiles in the Fig. 6(left) of the transition (a) $\mapsto$ (b)). On the other hand, irreversibility is a global constraint that changes the structure of the perturbation space, by only allowing positive perturbations in the damage field. Consequently, irreversible evolutions are qualitatively different from the unconstrained case.

The following example illustrates the subtle scenario in which two stable irreversible solutions are computed, one of which branches off the homogeneous solution at a bifurcation point without any instability. Despite allowing potential bifurcations, the second computation shows that the system can robustly navigate the purely homogeneous branch, satisfying a sufficient condition for stability (hence for observability) of the computed trajectory. This juxtaposition underscores a critical sensitivity in that the numerical bifurcation induced by the solver may not reflect an actual instability transition of the physical system, but rather an artifact of the computational method.

We compute an irreversible evolution using the same parameters as in the experiment in Figure 1(a) for the rigid substrate model, using two different numerical methods: first, the quasi-Newton L-BFGS/CG method (via ALGLIB) and second, a Newton solver [?] based on PETSc's reduced-space (active set) algorithms for variational inequalities (via SNES). In the first case, the numerical solution of first order optimality conditions with irreversibility constraints is tackled leveraging the minbleic subpackage of the Alglib library [12], via an active set method that handles inequality constraints in the first order equilibrium (or criticality) problem, encoded by inequality (20). Active inequality constraints correspond nodes

where damage necessarily evolves. Indeed, owing to the complementarity conditions (22).3, damage evolves ((22).1 holds with a strict inequality) in the regions where the system satisfies first order optimality conditions (22).2 with an equality. At first order, this is handled by projecting the energy gradient onto the subspace orthogonal to the set of active constraints, ensuring the solver proceeds in directions that respect irreversibility. This projection modifies the search space as the set of active constraints evolves at each load, requiring the algorithm to re-evaluate the target function, the constraints, and the constrained subspace at each variation of damage. This guarantees that the damage field respects the irreversibility condition throughout the simulation, ensuring robust and accurate enforcement of the first-order criticality, essential for the fidelity of phase-field models in fracture mechanics, introducing a significant computational overhead. Indeed, every time a constraint activates or deactivates (e.g., when a node reaches or leaves the energy optimality threshold), the constraint matrix is reorthogonalised. This operation has a computational cost of $O(n + \Delta k)k$, where n is the total number of degrees of freedom, k is the number of active constraints, and Δk represents the incremental changes in the active set. Furthermore, each evaluation of the target energy functional incurs an additional computational cost $O(n)$. By our choice of kinematic boundary conditions and the one-dimensional setting, the damage criterion is attained as soon as the load is nonzero throughout the whole computational domain hence this re-evaluation of constraints is performed for all nodes. ^{ALB} —

To solve the second-order cone-constrained inequality (27), both our solvers^{ALB} employ a numerical method based on the orthogonal decomposition of the Hilbert space X_0 according to two mutually polar cones K_0^+ and K^* , cf. [56]. Given an element z in X_0 , ~~it can be shown that~~^{ALB} there exists a unique decomposition into two orthogonal components, $x \in K_0^+$ and $y \in K^*$, where x and y are the closest elements to z in K_0^+ and K^* , respectively.

This decomposition allows us to project the second order problem into the cone and ensure that the eigen-solution satisfies the constraints imposed by irreversibility. This approach is particularly useful in mechanics and physics when dealing with unilateral constraints or problems where the solution space is naturally bounded by physical considerations (e.g., non-negative stress, plastic deformations, etc.)

More in particular, we implement a simple iterative Scaling-and-Projection algorithm [60] which depends upon one numerical parameter, a scaling factor $\eta > 0$, and an initial guess^{ALB}. Given a convex cone K and an initial guess z_0 (not necessarily in the cone), the algorithm operates by first projecting the vector in the cone, $x^{(k=0)} = P_K(z_0)$, then, the iterative estimate for the eigenvalue can be computed using the Rayleigh quotient

$$\lambda^{(k)} = \frac{x^{(k)T} H x^{(k)}}{\|x^{(k)}\|}, \quad (36)$$

where H is the (projected)^{ALB} Hessian operator. Then, we compute the residual vector $y^{(k)} = Hx^{(k)} - \lambda^{(k)}x^{(k)}$ and obtain the next iterate $x^{(k+1)} = v^{(k)}/\|v^{(k)}\|$ where $v^{(k)} = P_K(x^{(k)} + \eta y^{(k)})$. The algorithm is repeated until convergence is achieved on $x^{(k)}$ ^{ALB}. Note that, in the cone-constrained case, the residual vector $y^{(k)}$ need not be zero at convergence.

In Figure 9.left, we show the evolution of the damage field under the irreversibility constraint. Corresponding spectra are displayed in Figure 9.right where coloured markers represent a stable solution that bifurcates, computed with ALGLIB/L-BFGS, and circle-marked eigenvalues correspond to a uniformly stable homogeneous solution computed with PETSc/SNES. Starting from zero damage at zero load, damage uniformly increases as the solution evolves along the homogeneous branch, matching the theoretical prediction that the evolution is unique below the critical (bifurcation) load. As soon as the load approaches this threshold, a small oscillation emerges before the theoretical bifurcation point, reminiscent of the eigenmode ($n = 3$) computed by Fourier series, cf. Fig 6(a). Figure 9.left shows the profiles of the damage field at different loads, highlighting the profile at the theoretical bifurcation load (plotted in red), corresponding to the loss of uniqueness of the evolution. A small perturbation is observed in the damage field before the bifurcation point, which is not consistent with the theoretical predictions, highlighted in the inset showing the L^2 norm of the damage gradient. Following loading, the numerical solver switches from the homogeneous solution to a localised pattern. The corresponding trajectory is shown in the equilibrium map in Figure 2. The irreversible homogeneous and localised branches, both stable, are displayed with black dashed and black solid lines, respectively. Figure 9.right displays the irreversible constrained spectra associated to both evolutions, informing on the uniqueness and stability of the corresponding solution trajectories. Both solvers agree on the critical load for bifurcation, yet differ in the computed evolution path, reinforcing the need for thorough stability and bifurcation analyses in the irreversible case and the choice of a solution strategy that is robust and efficient in not inducing purely numerical instabilities. Indeed, the L-BFGS/CG method shifts the homogeneous solution to a

distinct trajectory that diverges from the homogeneous path even before the bifurcation load, despite the stability of the homogeneous branch thus exhibiting a spurious bifurcation not associated with a stability transition. Contrarily, the second solver correctly identifies the loss of uniqueness of the homogeneous solution, and remains on the homogeneous branch showing continuous (positive) cone-eigenvalues across the bifurcation point. Indeed, the homogeneous solution should remain observable until, if ever, loss of stability is reached. Note the infimum spectrum obtained for the bifurcation problem is piecewise continuous, indicating smooth variation in the eigenvalue as the load increases. Conversely, the cone spectrum differs from the ball spectrum in that it is piecewise smooth but globally discontinuous. Our numerical experience hints at the sensitivity of the cone-constrained eigenvalue problem to the initial guess. This can be seen in the bifurcation spectra for small loads, where the first eigenmode of the bifurcation problem -used to initialise the cone-solver- has a definite sign. In these conditions, when the stability solver is initialised with an initial guess that is orthogonal to the cone (i.e. negative definite), the Scaling-and-Projection algorithm effectively starts from a zero initial condition. This turns out to be suboptimal, leading to an over estimation of the constrained eigenvalue. This can be observed in branches a and b in Figure 9.right at small loads. At higher loads on the contrary, the initial condition is a nontrivial perturbation mode close to the optimal constrained mode, leading to a more accurate estimation of the stability threshold associated to the infimum of the constrained spectrum. In Fig. 8.left (resp., right) we plot the damage state α and corresponding bifurcation eigenmode p (associated to point A in Figure 9.right) and stability eigenmodes p^+ (associated to point B in 9.right) at the bifurcation loading step $\bar{\epsilon}_b$, namely point H in the energy diagram Fig. 2. This analysis highlights the delicate interplay between numerical methods, the variational structure of the problem, and the underlying physical systems, emphasising that observed numerical bifurcations under these conditions may be interpreted with caution to avoid potential numerical artifacts. ^{ALB}

Finally, note that the bifurcation spectrum is singular at $\bar{\epsilon}_t = 0$ following the fact that, in our model, the damage criterion is attained as soon as the load is non-zero, thus the space of admissible rate perturbations changes suddenly from $H_0^1(0, 1) \times \emptyset$ at $\bar{\epsilon}_t = 0$ to the full space $X_0 = H_0^1(0, 1) \times H^1(0, 1)$ for $\bar{\epsilon}_t > 0$ which includes all (sufficiently smooth) rate perturbations. Conversely, because the set of admissible state perturbations is strictly included in the set of admissible rates, the stability eigenvalues are equal or larger than the unconstrained ones, they coinciding only if unconstrained modes have a definite sign. In the illustrative example at hand, despite the occurrence of negative eigenvalues for the bifurcation problem, the eigenvalues of the stability problem are all positive, which is a sufficient condition to ascertain the stability (and thus, the observability) of the computed evolution.

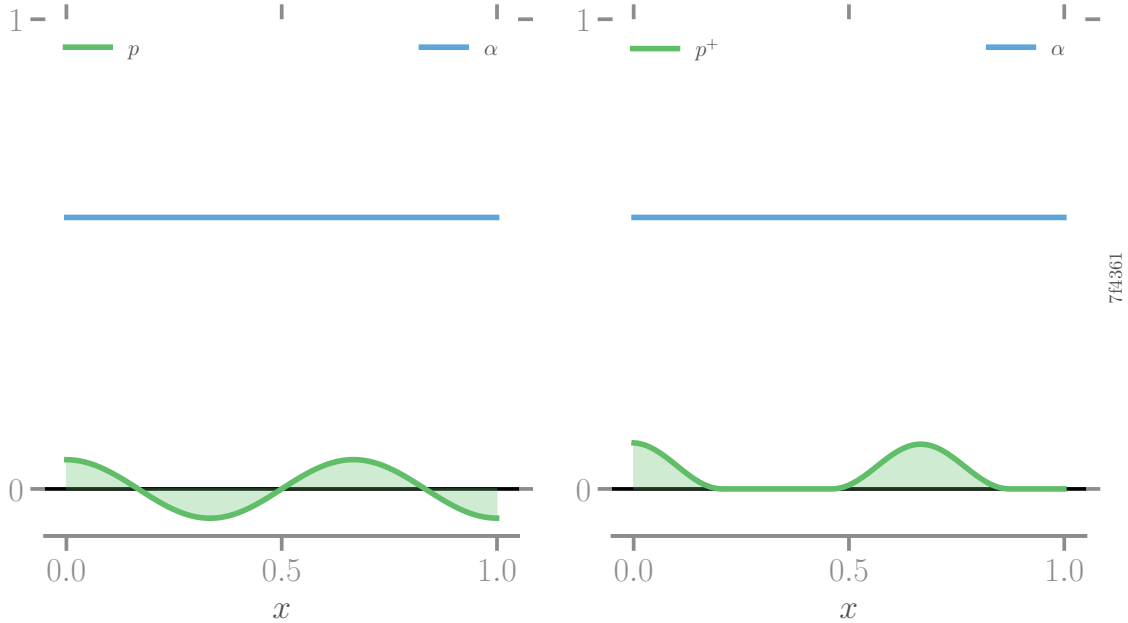


Figure 8: Damage field profiles (blue) and inf-eigenvectors (green) at bifurcation load $\bar{\epsilon}_b$ for bifurcation problem (left) and stability problem (right). At bifurcation, damage is homogeneous. Bifurcation mode is eigenmode on branch $n = 4$ (cf. Fig. 2-right). Stability problem inf-eigenvector (right, green) corresponds to $\lambda_b = 2 \cdot 10^{-2}$. ^{OUS}

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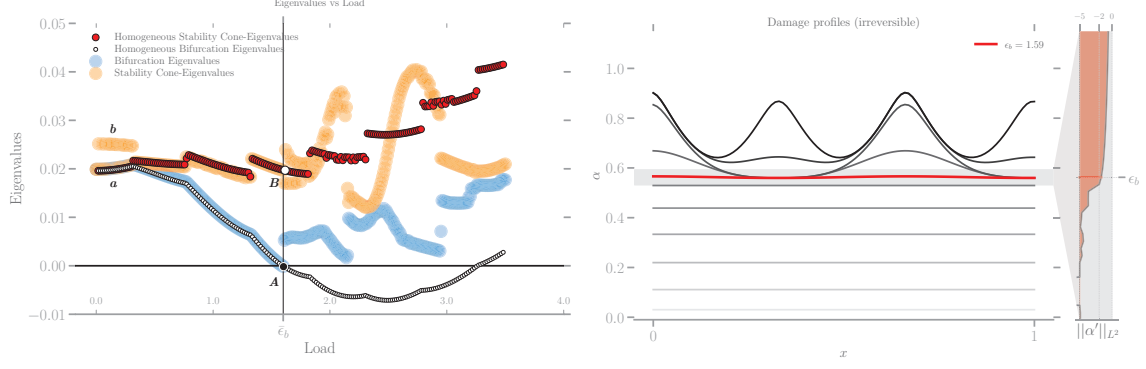


Figure 9: Left: Hessian spectra lower bounds for bifurcation (orange line) and stability problems (blue circles) along homogeneous evolution. Evolution path remains unique until $\bar{\epsilon}_b$, after which homogeneous solution stays stable despite multiple solutions existing, preventing crack nucleation with active irreversibility constraint. Note λ_t (inf-eigenvalue cone) discontinuity. Right: Quasi-Newton solver solutions showing slow damage evolution due to numerical errors.^{OUS}

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5 Discussion

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Capturing branch switching phenomena across stability transitions is not an automatic feature of approximate numerical methods. If they rely on approximate information about the Hessian of the energy functional, these methods do not guarantee to systematically detect transitions between critical equilibrium states, when stability is lost. Indeed, this requires a careful determination of the zero-eigenmodes that render singular the exact nonlinear Hessian, which is typically not available in general purpose first order numerical algorithms. In practice, without such information, critical loads for equilibrium transitions become algorithm-dependant and are not consistent with closed-form solutions of the exact evolution problem.

Our evidence is that certain numerical methods can introduce non-physical artifacts which should be distinguished from genuine physical phenomena. Our ongoing work aims to refine numerical techniques to provide more reliable algorithms for analysing irreversible processes in variational evolutionary problems with multiple local minima and a high number of degrees of freedom.

From our numerical experience, first order solutions to strongly nonlinear, nonconvex, and singular problems, like those of interest in the applications exhibit strong sensitivity to numerical errors, possibly leading to spurious bifurcations and artificial state transitions. On the other hand, solutions which integrate second order information are robust and their observability can be fully characterised.

More than numerical perturbations (which can always arise,) the use of numerical methods relying only on (conjugate) gradients (in lieu of exact Hessians) is prone to introducing non-physical crack nucleation. Our analysis shows that while quasi-Newton algorithms accelerate convergence, this comes at the cost of introducing unphysical effects. In particular, the approximations inherent in quasi-Newton updates may lead to solutions that appear physical yet are driven by numerical disorder rather than the governing equations. This sensitivity indicates that when irreversibility constraints are critical, a full Hessian approach is often necessary to accurately capture the stability and bifurcation behavior. Ultimately, practitioners must carefully balance the computational efficiency of quasi-Newton methods against the increased reliability (and potential cost) of employing the full Hessian.^{ALB} This is an important observation, which highlights the need for a thorough investigation of the stability of solutions. If only physical factors are considered, an energetic selection mechanism is already encapsulated in the stability statement in the evolution law. As a consequence, equilibrium solutions under increasing load should be maintained as observable if stable, assuming that no nucleation should occur otherwise.

We present two main options for discussing these considerations in view of the importance of using accurate and robust algorithms in real scenarios:

1. Ignore the Numerical Artifact: focussing solely on first order considerations and acknowledging that the observed computed nucleations may be purely numerical and should not be considered in physical terms.

2. Highlight the Numerical Artifact: Alternatively, emphasizing that homogeneous solutions should be observable, despite the sensitivity to numerical parameters (artifacts, in the quasi-Newton approach) and the abundance of admissible solutions (in the nonconvex scenario), or
3. Otherwise.

Suggesting that state transitions in complex scenarios should be carefully interpreted, the connection between observability and stability is functional to understanding real patterns that emerge, e.g., in higher dimensions or in other physical systems.

In either case, our computations show that, unless second order analysis is performed, observed nucleations are not *necessarily* indicative of physical cracks but rather of an interplay between purely physical phenomena, inherent to the nature of natural processes, and numerical biases inherent to the computational methods employed. This distinction is crucial for understanding the limitations and proper application of numerical techniques as predictive tools in contexts where cracks are a real concern for structures.

Future work will include a more detailed exploration of evolutionary algorithms and their implementation for stability analysis of fracture in thin films. Some notable instances are craquelures in artistic paintings [28, 13, 14], brittle stability of the cryosphere [82, 78, 76, 55] and crack-pattern selection in metallic thin films [26]. As understood in this work, the final crack patterns depend heavily on the form of the selected unstable modes. In turn, the irreversibility constraint, primarily implemented to prevent crack closure, directly impacts the stability of solutions and mode selection at transition loads. Therefore, our findings suggest that the success of the phase-field model in predicting crack patterns in thin films relies on the use of robust numerical algorithms.

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